2026 Fall

CS270: Digital Image Processing

Register here to download lecture slides !

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Lecture 5
Frequency Domain - Foundations

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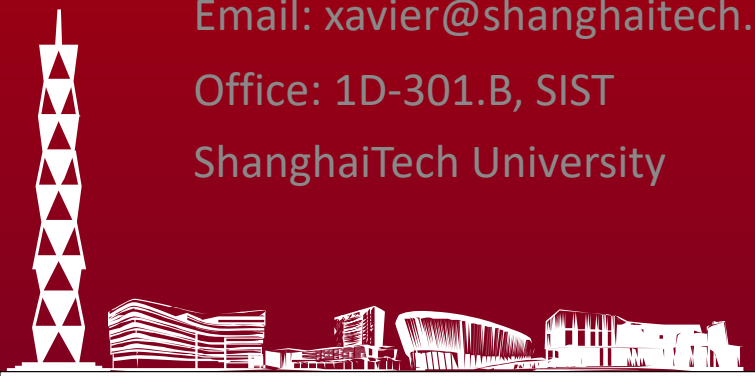

2026 Fall – Oct 14

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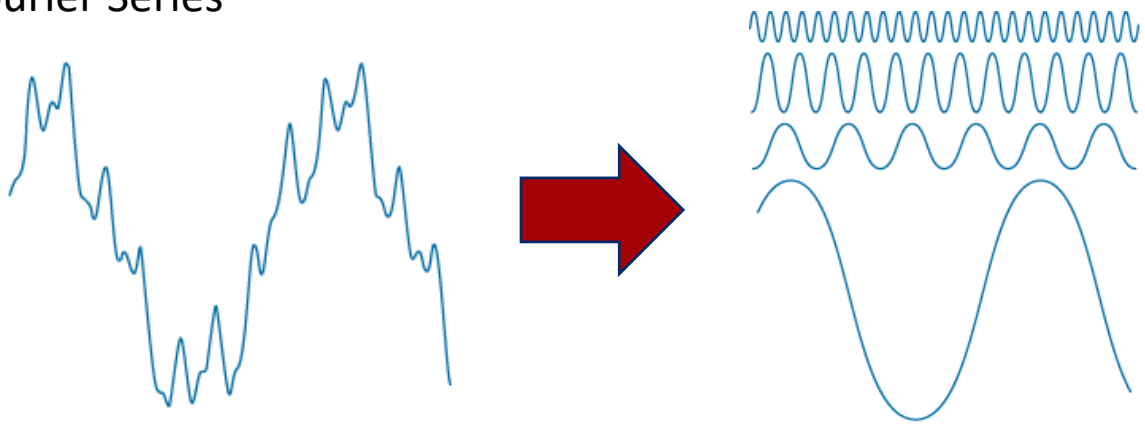
Outline

- Why Fourier Transform in DIP
- Introduction (recap) of Fourier Transform (CTFT, DTFT, DFT)
- Discrete Fourier Transform and its Inverse (1D & 2D)
- Sampling Theorem & Aliasing (1D & 2D)
- Properties of 2D DFT and IDFT
- 2D DFT implementation

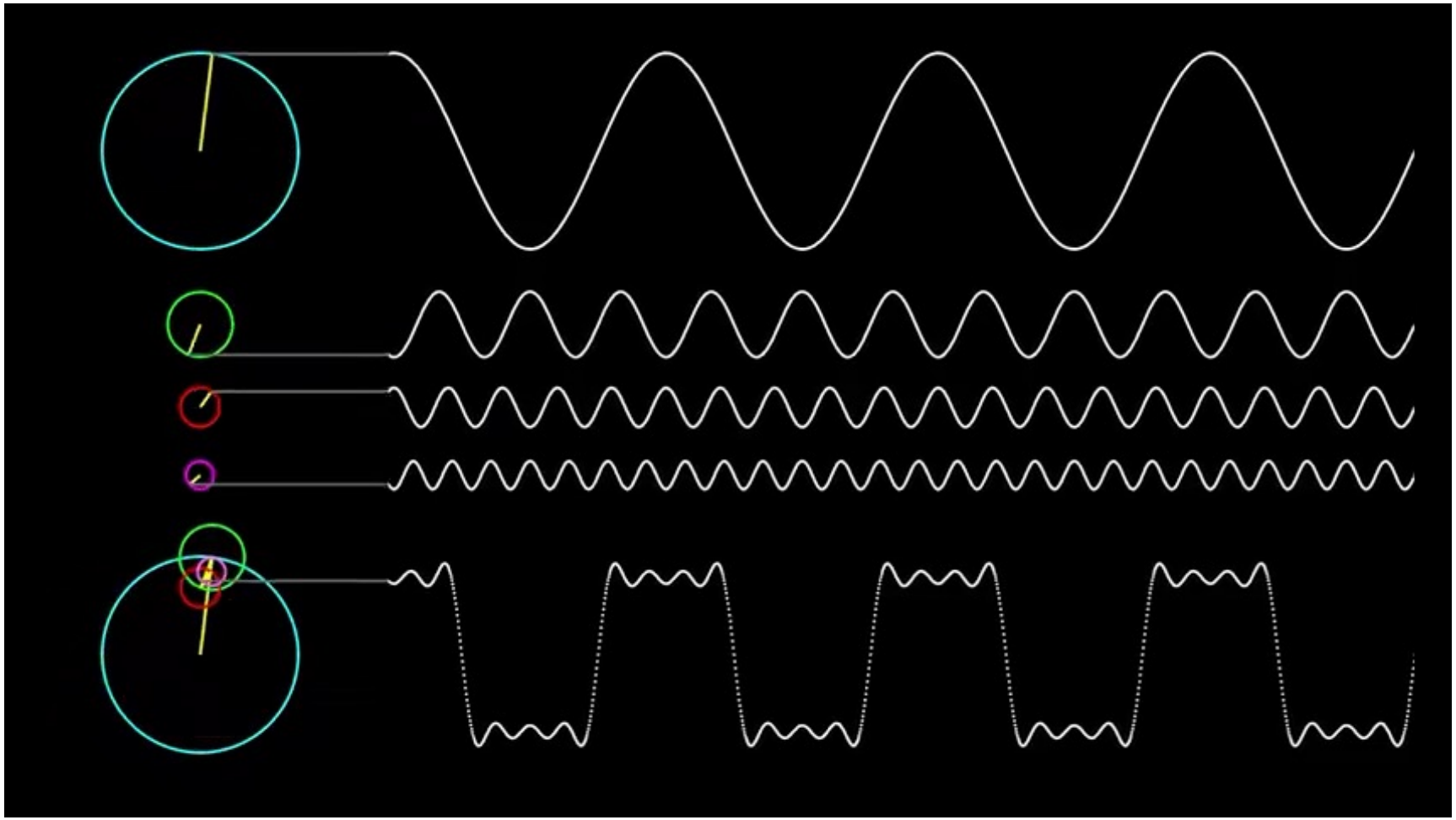
Jean Baptiste Joseph Fourier (1768-1830)

- French mathematician
- Had a crazy idea (1807)
  - Any periodic function can be rewritten as a weighted sum of sines and cosines of different frequencies
- Who didn't believe it?
  - Lagrange, Laplace, Poisson…
  - Not translated into English until 1878!
- But it's true!
  - We call it Fourier Series



Fourier Series

- https://www.youtube.com/watch?v=cUD1gMAl6W4



Why Fourier Transform in DIP

- Exposing image features not visible in spatial domain
- Image filtering in frequency domain (sometimes even faster!)
- Image compression or encoding
- Image reconstruction or recovery
- Classification
- …
- Or simply to provide another perspective to look at the image

Why Fourier Transform in DIP

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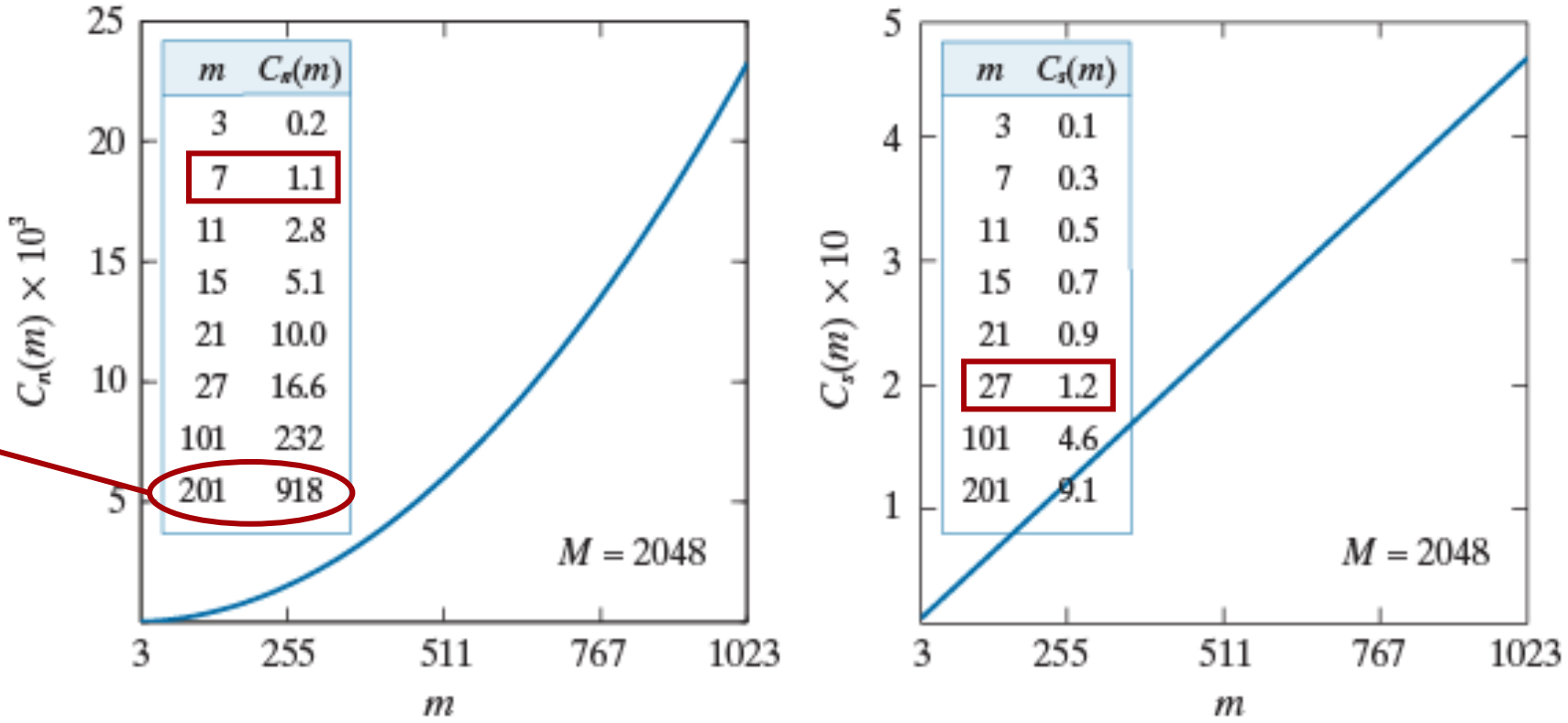
- **The importance of FFT: Computational advantage $C(m)$**

1min with FFT = 17h with correlation !



m	$C_n(m)$
3	0.2
7	1.1
11	2.8
15	5.1
21	10.0
27	16.6
101	232
201	918

m	$C_s(m)$
3	0.1
7	0.3
11	0.5
15	0.7
21	0.9
27	1.2
101	4.6
201	9.1

a b

FIGURE 4.2 (a) Computational advantage of the FFT over non-separable spatial kernels. (b) Advantage over separable kernels. The numbers for $C(m)$ in the inset tables are not to be multiplied by the factors of 10 shown for the curves.

If $C(m) > 1$, FFT is faster

Introduction (recap) of Fourier Transform

- **By using Euler's Formula**

$$e^{j\theta} = \cos\theta + j.\sin\theta$$

- **Fourier series for periodic functions**

$$f(t) = \sum_{n=-\infty}^{\infty} c_n . e^{j.\frac{2\pi n}{T}t}$$

$$c_n = \frac{1}{T}\int_{-T/2}^{T/2} f(t)e^{-j.\frac{2\pi n}{T}t}\,dt, \qquad n = 0, \pm 1, \pm 2, \ldots$$

Introduction (recap) of Fourier Transform

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- **Fourier transform:**

$$\mathcal{F}[f(t)] = F(\mu) = \int_{-\infty}^{\infty} f(t)e^{-j.2\pi\mu t}\,dt$$

$$F(\mu) = \int_{-\infty}^{\infty} f(t)[\cos(2\pi\mu t) - j\sin(2\pi\mu t)]\,dt$$

Frequency

- **Inverse Fourier transform**

$$f(t) = \mathcal{F}^{-1}[F(\mu)] = \int_{-\infty}^{\infty} F(\mu)e^{j.2\pi\mu t}\,d\mu$$

Fourier Transform: Example 1 - Box

$$F(\mu) = \int_{-\infty}^{\infty} f(t)e^{-j.2\pi\mu t}\,dt = \int_{-W/2}^{W/2} Ae^{-j.2\pi\mu t}\,dt$$

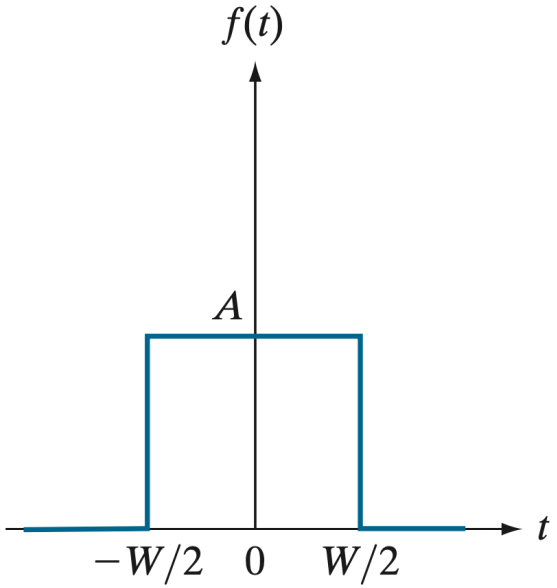
$$= \frac{-A}{j2\pi\mu}\left[e^{-j2\pi\mu t}\right]_{-W/2}^{W/2} = \frac{-A}{j2\pi\mu}\left[e^{-j\pi\mu W} - e^{j\pi\mu W}\right]$$

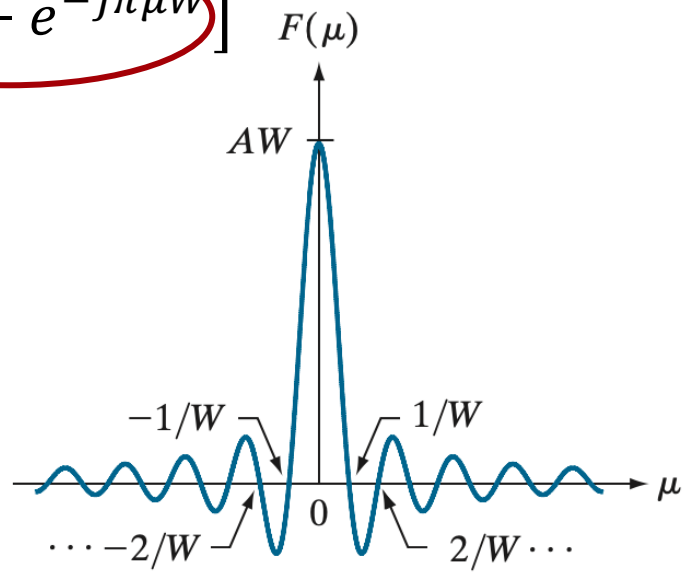
$$= \frac{A}{j2\pi\mu}\left[e^{j\pi\mu W} - e^{-j\pi\mu W}\right]$$

$$= AW\frac{\sin(\pi\mu W)}{\pi\mu W}$$

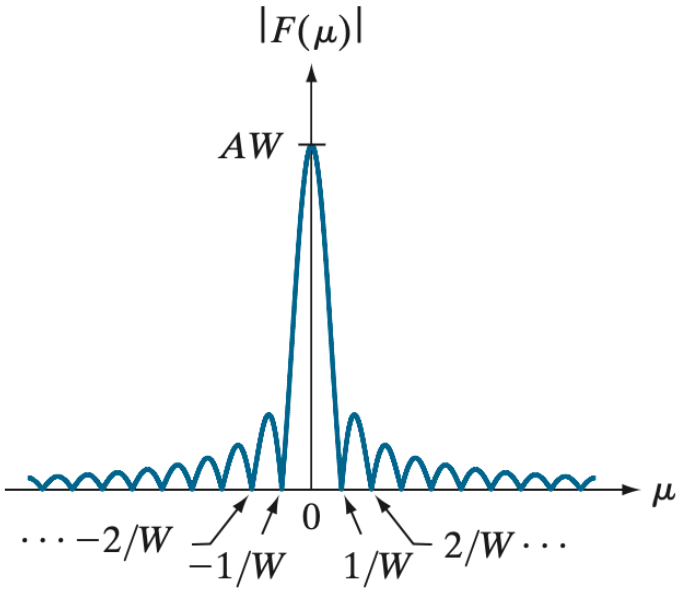
$$= AW\,\mathrm{sinc}\,\mu W$$

$$\sin\theta = (e^{j\theta} - e^{-j\theta})/2j$$





Fourier Transform

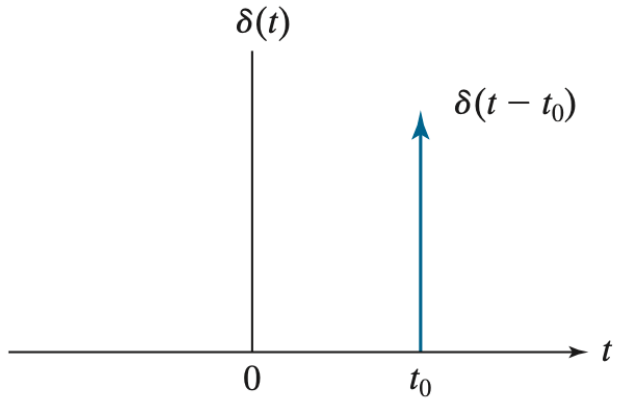


Spectrum (magnitude of FT)

Continuous Impulses

- **Unit impulse**

$$\delta(t) = \begin{cases} \infty & \text{if } t = 0 \\ 0 & \text{if } t \neq 0 \end{cases} \qquad \int_{-\infty}^{\infty} \delta(t)\,dt = 1$$

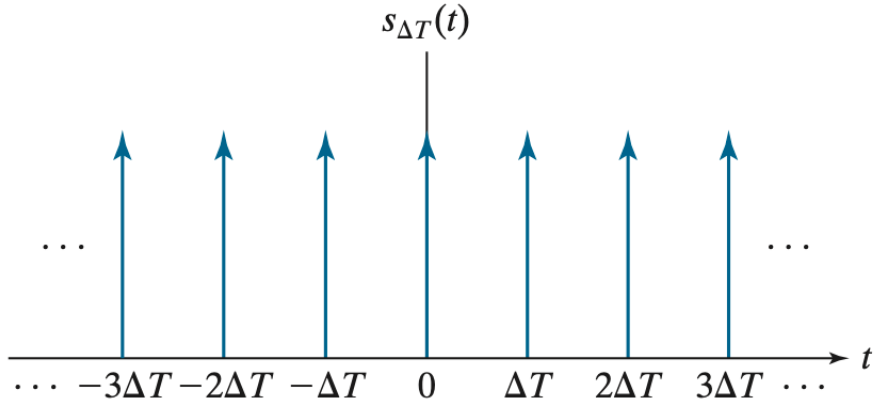


- **Sifting property**

$$\int_{-\infty}^{\infty} f(t)\delta(t)\,dt = f(0) \qquad \int_{-\infty}^{\infty} f(t)\delta(t - t_0)\,dt = f(t_0)$$

- **Impulse train**

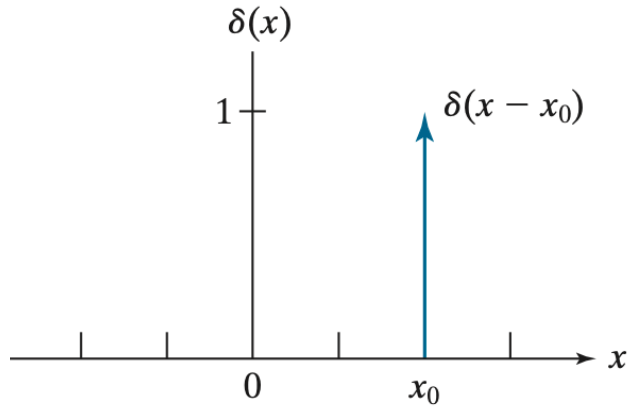
$$s_{\Delta T}(t) = \sum_{k=-\infty}^{\infty} \delta(t - k\Delta T)$$



Discrete Impulses

- **Unit impulse**

$$\delta(x) = \begin{cases} 1 & \text{if } x = 0 \\ 0 & \text{if } x \neq 0 \end{cases} \qquad \sum_{x=-\infty}^{\infty} \delta(x) = 1$$

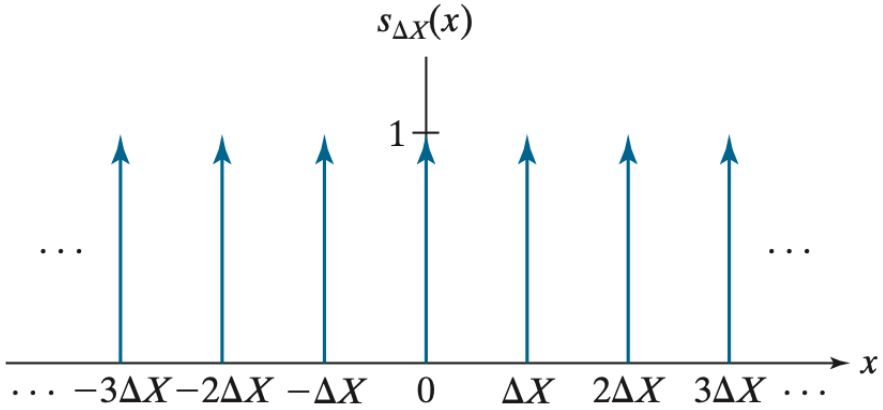


- **Sifting property**

$$\sum_{x=-\infty}^{\infty} f(x)\delta(x) = f(0) \qquad \sum_{x=-\infty}^{\infty} f(x)\delta(x - x_0) = f(x_0)$$

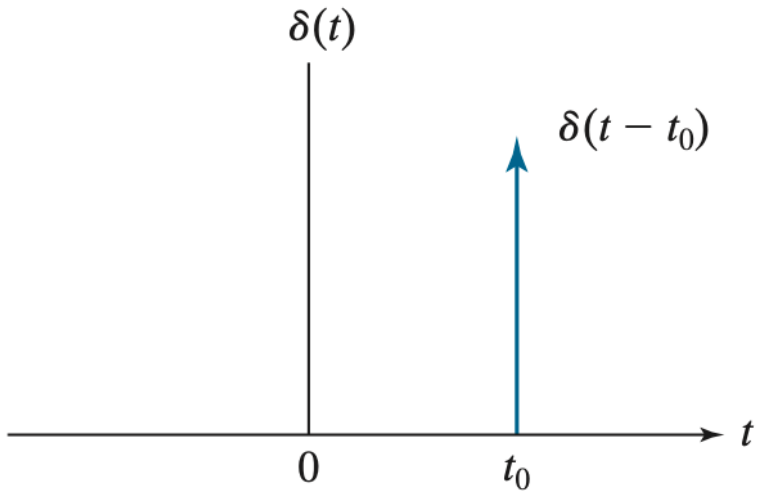
- **Impulse train**

$$s_{\Delta T}(x) = \sum_{k=-\infty}^{\infty} \delta(x - k\Delta X)$$



Fourier Transform: Example 2 - Impulse

$$\mathcal{F}[\delta(t)] = F(\mu) = \int_{-\infty}^{\infty} \delta(t) e^{-j.2\pi\mu t}\, dt = 1$$



$$\mathcal{F}[\delta(t - t_0)] = F(\mu) = \int_{-\infty}^{\infty} \delta(t - t_0) e^{-j.2\pi\mu t}\, dt = e^{-j.2\pi\mu t_0}$$

Fourier Transform: Example 3 – Impulse train

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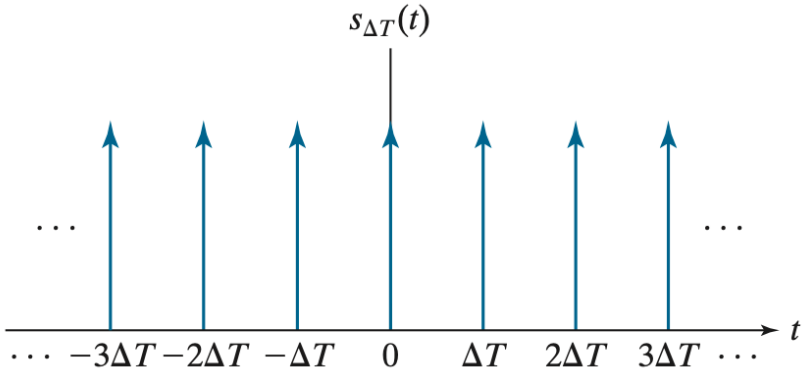
$$f(t) = \mathcal{F}^{-1}[F(\mu)] = \int_{-\infty}^{\infty} F(\mu) e^{j.2\pi\mu t} d\mu$$

$$\mathcal{F}[f(t)] = F(\mu) = \int_{-\infty}^{\infty} f(t) e^{-j.2\pi\mu t} dt$$

$$\mathcal{F}[F(t)] = f(-\mu)$$



➡ Since $\mathcal{F}[\delta(t - t_0)] = e^{-j.2\pi\mu t_0}$ then $\mathcal{F}\left[e^{-j.2\pi t t_0}\right] = \delta(-\mu - t_0)$

➡ By using $-t_0 = a$ then $\mathcal{F}\left[e^{j.2\pi a t}\right] = \delta(-\mu + a) = \delta(\mu - a)$

$s_{\Delta T}(t)$ has period ΔT so, we can express it as a Fourier series:

$$s_{\Delta T}(t) = \sum_{n=-\infty}^{\infty} c_n e^{j\frac{2\pi n}{\Delta T}t}$$

$$c_n = \frac{1}{\Delta T}\int_{-\Delta T/2}^{\Delta T/2} s_{\Delta T}(t) e^{-j\frac{2\pi n}{\Delta T}t} dt$$

Only contains pulse at t=0 !

$$c_n = \frac{1}{\Delta T}\int_{-\Delta T/2}^{\Delta T/2} \delta(t) e^{-j\frac{2\pi n}{\Delta T}t} dt = \frac{1}{\Delta T} e^0 = \frac{1}{\Delta T}$$

➡ $$s_{\Delta T}(t) = \frac{1}{\Delta T}\sum_{n=-\infty}^{\infty} e^{j\frac{2\pi n}{\Delta T}t}$$

Since we showed that $\mathcal{F}\left[e^{j\frac{2\pi n}{\Delta T}t}\right] = \delta\left(\mu - \frac{n}{\Delta T}\right)$

$$\mathcal{F}[s_{\Delta T}(t)] = S(\mu) = \mathcal{F}\left[\frac{1}{\Delta T}\sum_{n=-\infty}^{\infty} e^{j\frac{2\pi n}{\Delta T}t}\right] = \frac{1}{\Delta T}\sum_{n=-\infty}^{\infty} \delta\left(\mu - \frac{n}{\Delta T}\right)$$

Fourier Transform: Convolution

$$(f \star h)(t) = \int_{-\infty}^{\infty} f(\tau)h(t-\tau)\,d\tau$$

$$\mathcal{F}[(f \star h)(t)] = \int_{-\infty}^{\infty}\left[\int_{-\infty}^{\infty} f(\tau)h(t-\tau)\,d\tau\right]e^{-j.2\pi\mu t}dt$$

$$= \int_{-\infty}^{\infty} f(\tau)\left[\int_{-\infty}^{\infty} h(t-\tau)e^{-j.2\pi\mu t}\,dt\right]d\tau$$

$$\mathcal{F}[h(t-\tau)] = H(\mu)e^{-j.2\pi\mu\tau}$$

$$= \int_{-\infty}^{\infty} f(\tau)\left[H(\mu)e^{-j.2\pi\mu\tau}\right]d\tau$$

$$= H(\mu)\int_{-\infty}^{\infty} f(\tau)e^{-j.2\pi\mu\tau}d\tau$$

$$= H(\mu)F(\mu)$$

$$\mathcal{F}[(f \star h)(t)] = (H \cdot F)(\mu)$$

$$(f \star h)(t) \Leftrightarrow (H \bullet F)(\mu)$$

$$(f \bullet h)(t) \Leftrightarrow (H \star F)(\mu)$$

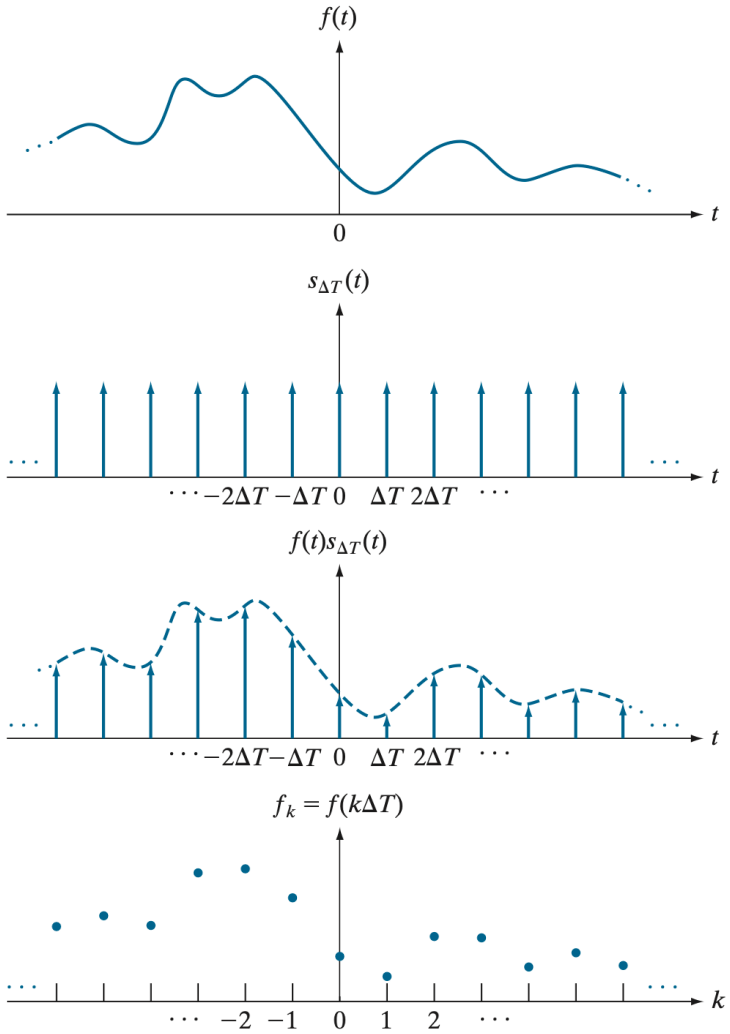
A look back on sampling

- Let's consider a continuous function $f(t)$ that we sample at uniform intervals ΔT

$$\tilde{f}(t) = f(t)s_{\Delta T}(t) = \sum_{n=-\infty}^{\infty} f(t)\delta(t - n\Delta T)$$

- Values of samples are given by

$$f_k = \int_{-\infty}^{\infty} f(t)\delta(t - k\Delta T)dt$$
$$= f(k\Delta T)$$



Fourier Transform of a sampled function

$$\tilde{F}(\mu) = \mathcal{F}\left[\tilde{f}(t)\right] = \mathcal{F}[f(t).s_{\Delta T}(t)]$$
$$= (F \star S)(\mu)$$

where $S(\mu) = \frac{1}{\Delta T}\sum_{n=-\infty}^{\infty} \delta\left(\mu - \frac{n}{\Delta T}\right)$

$$\tilde{F}(\mu) = (F \star S)(\mu) = \int_{-\infty}^{\infty} F(\tau)S(\mu-\tau)d\tau$$

$$= \frac{1}{\Delta T}\int_{-\infty}^{\infty} F(\tau)\sum_{n=-\infty}^{\infty}\delta\left(\mu-\tau-\frac{n}{\Delta T}\right)d\tau$$

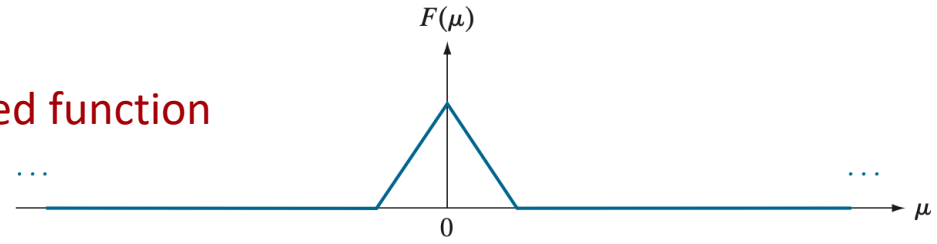
$$= \frac{1}{\Delta T}\sum_{n=-\infty}^{\infty}\int_{-\infty}^{\infty} F(\tau)\delta\left(\mu-\tau-\frac{n}{\Delta T}\right)d\tau$$

$$= \frac{1}{\Delta T}\sum_{n=-\infty}^{\infty} F\left(\mu-\frac{n}{\Delta T}\right)$$

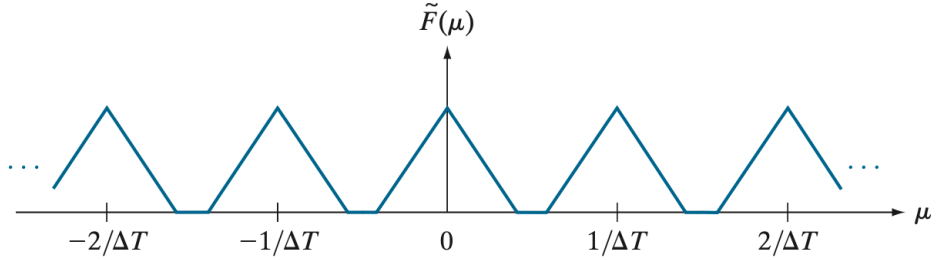
- *infinite*, *periodic* sequence of *copies* of the transform of the original, continuous function
- Separation driven by $^1/_{\Delta T}$
- $\tilde{f}(t)$ is sampled but $\tilde{F}(\mu)$ is a continuous function !

The Sampling Theorem

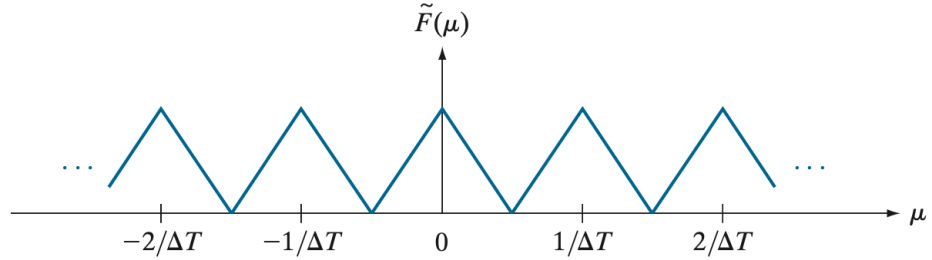
Fourier transform of a band limited function



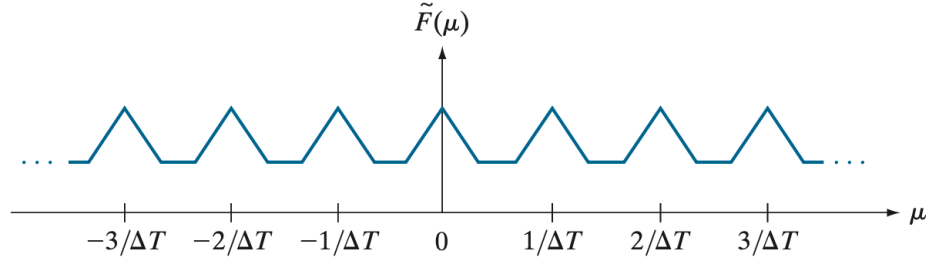
Over-sampling



Critically sampling



Under sampling

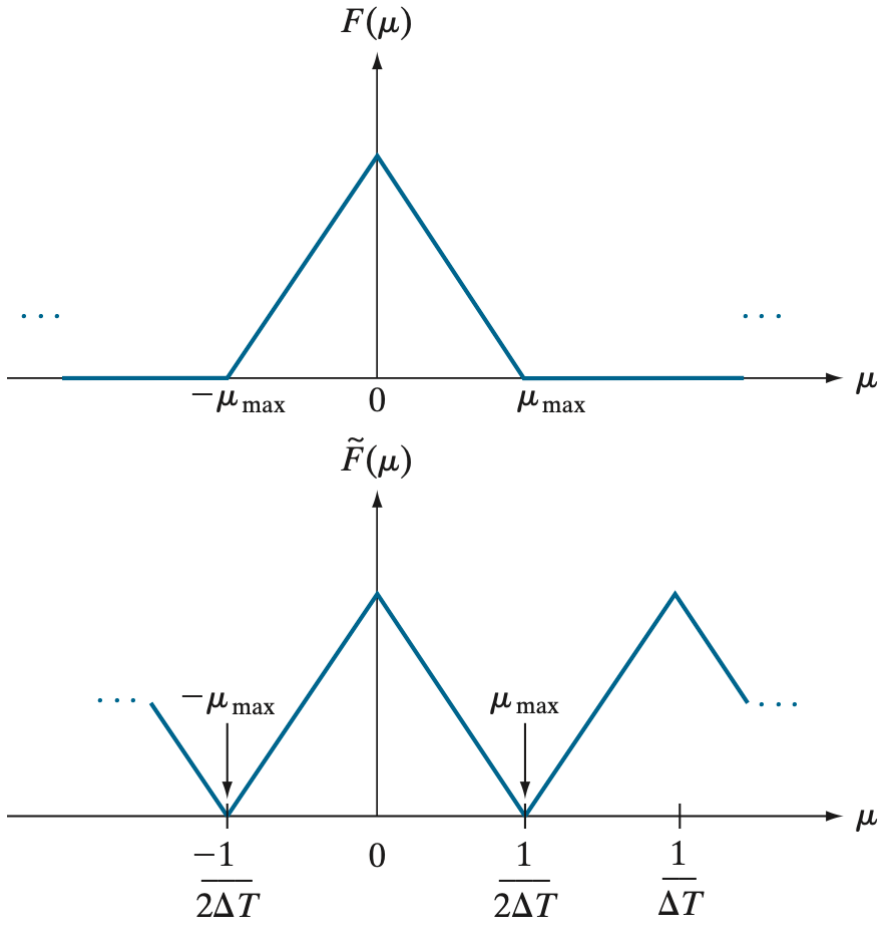


- Band limited function (带限函数) : Fourier is transform is 0 outside of a finite interval (band)
- $f(t)$ can be recovered if we can isolate a single copy of $F(\mu)$
- Need enough separation between the copies

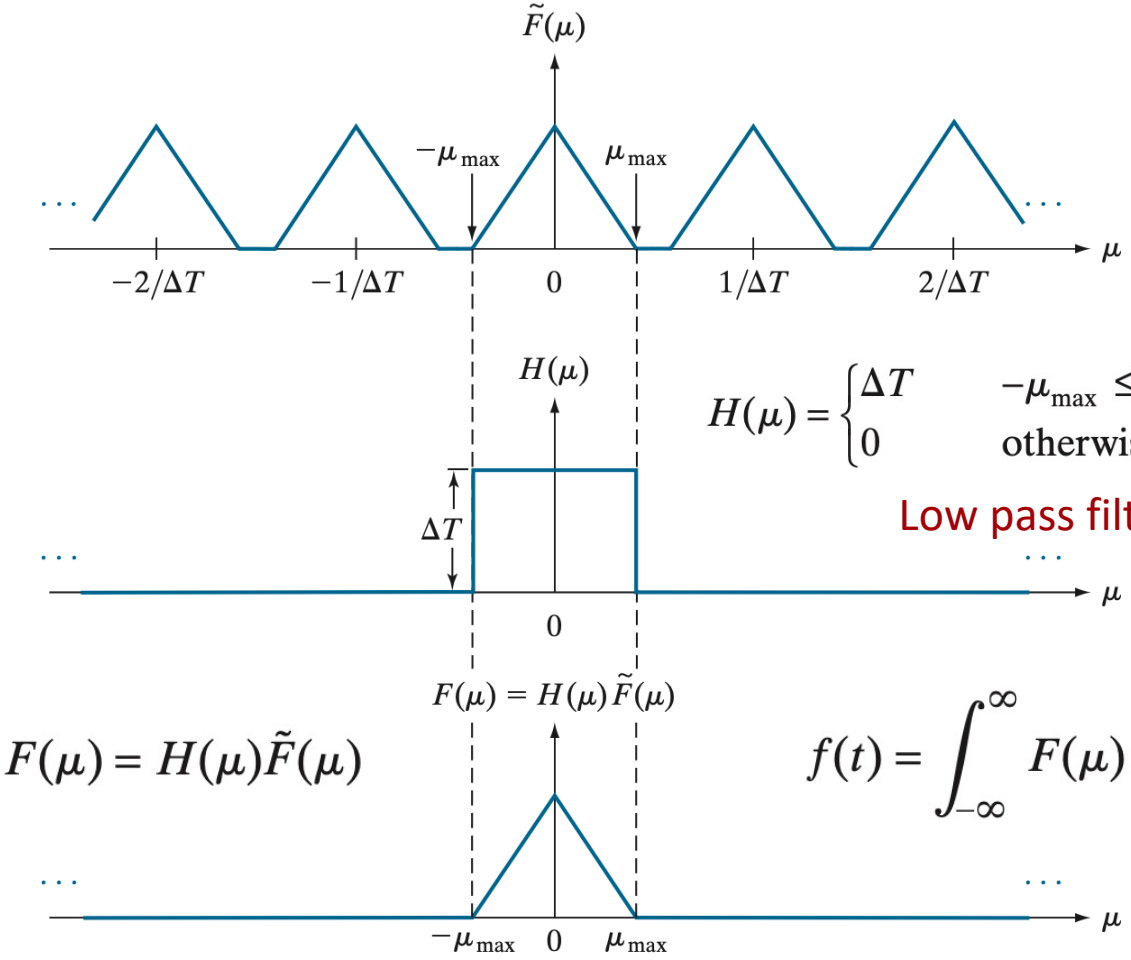
$$\frac{1}{\Delta T} > 2.\mu_{max}$$

The Sampling Theorem

Critically-sampled: $\frac{1}{\Delta T} = 2.\mu_{max}$



The Sampling Theorem - Reconstruction



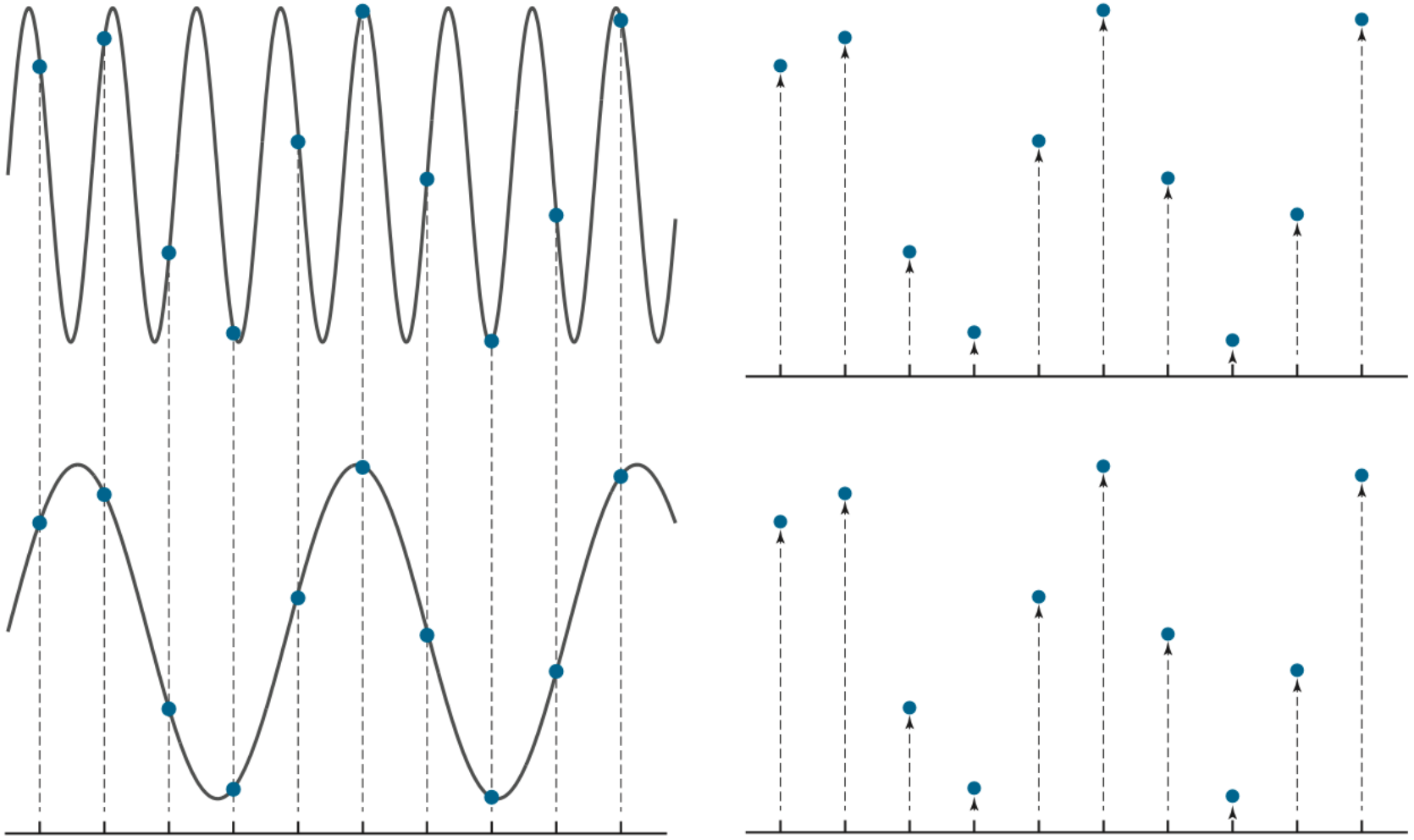
$$H(\mu) = \begin{cases} \Delta T & -\mu_{\max} \le \mu \le \mu_{\max} \\ 0 & \text{otherwise} \end{cases}$$

Low pass filter !

$$F(\mu) = H(\mu)\tilde{F}(\mu)$$

$$f(t) = \int_{-\infty}^{\infty} F(\mu) e^{j2\pi\mu t} d\mu$$

Aliasing (混叠)



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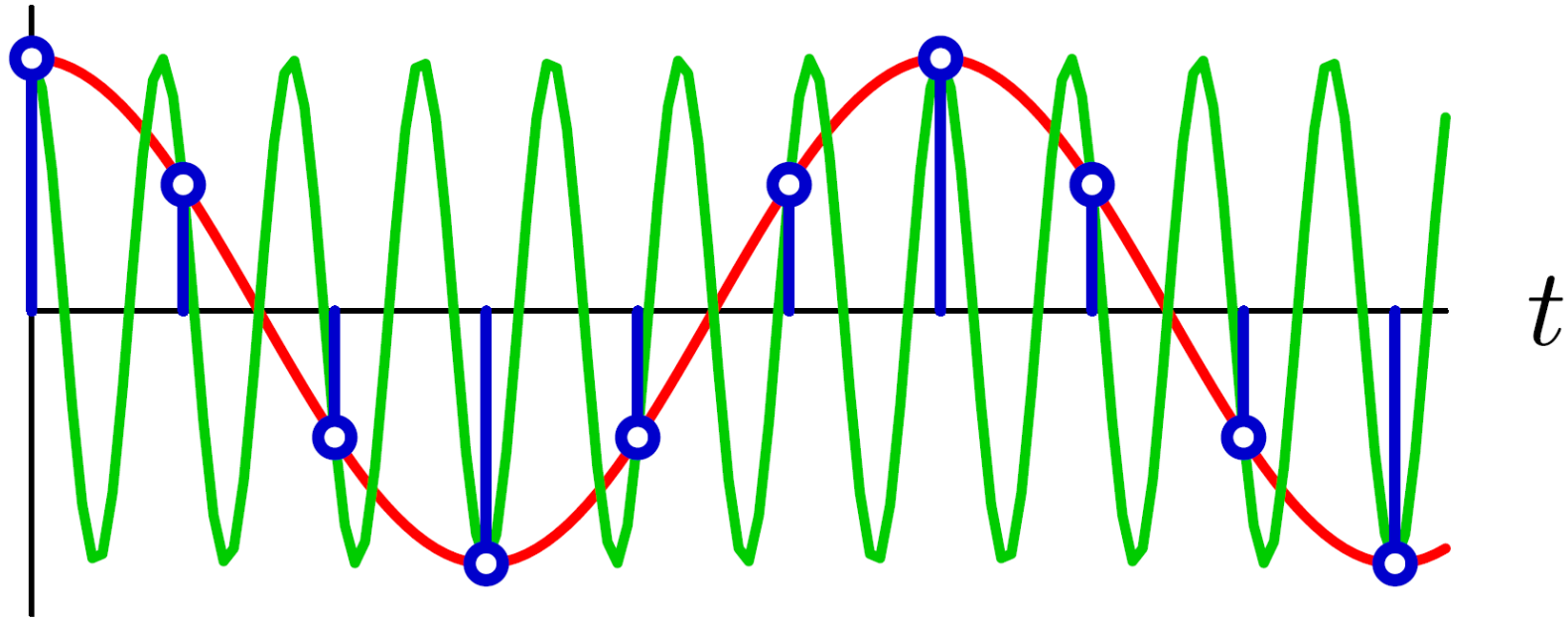
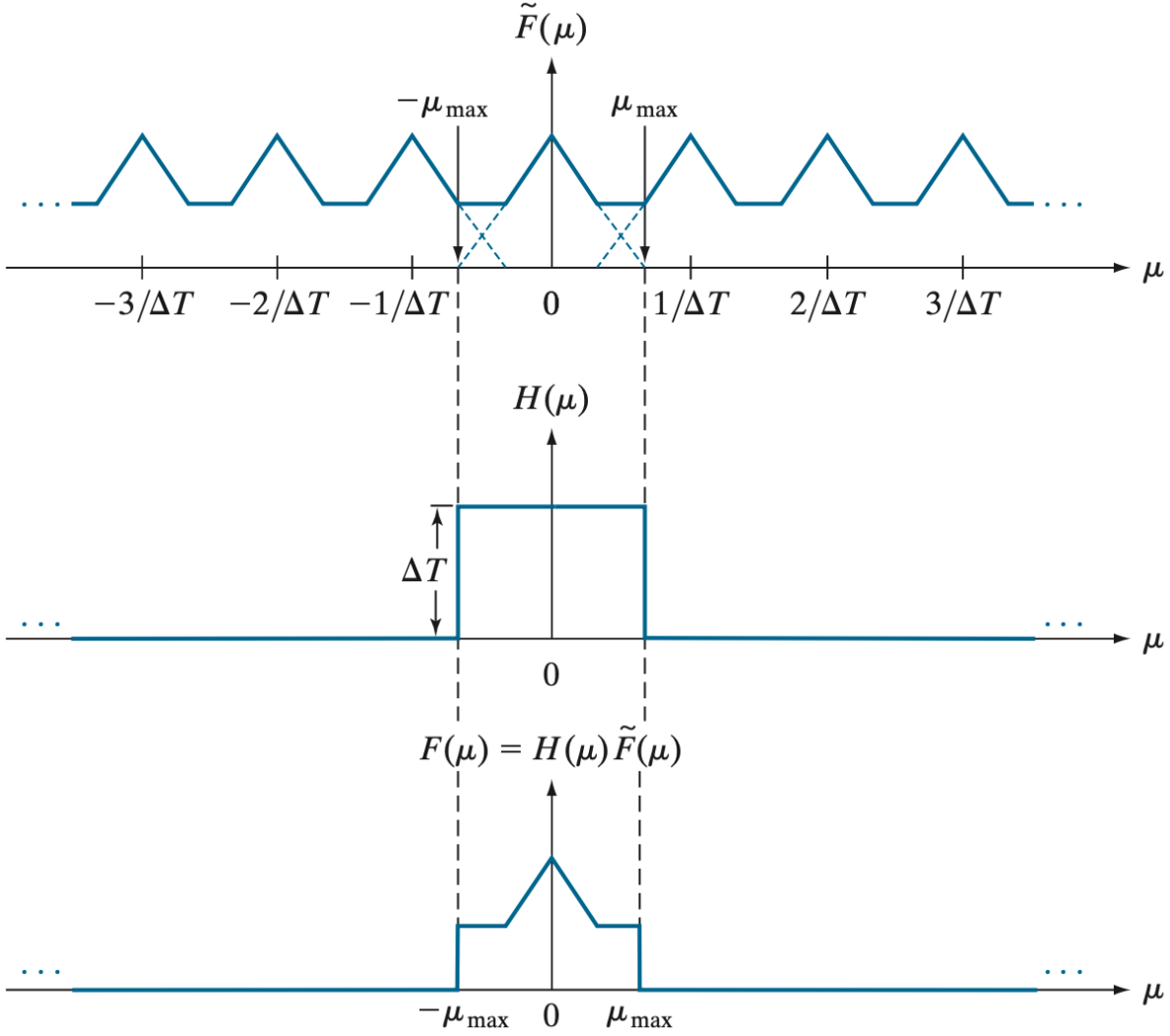
Aliasing (混叠)

$\cos \frac{7\pi}{3} n ?$ $\cos \frac{\pi}{3} n ?$



Aliasing and under-sampling



Limited-duration functions

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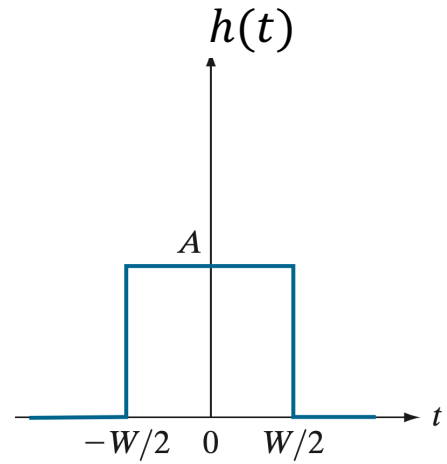
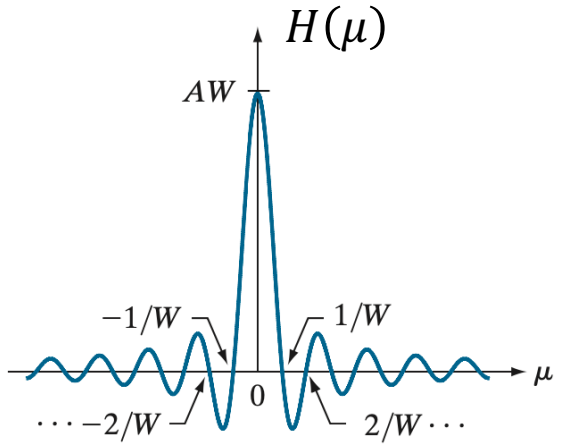
- Let's observe (i.e. limit) a band-limited function $f(t)$ over a finite interval $[0, T]$
- Equivalent to multiplying by

$$h(t) = \begin{cases} 1 & 0 \le t \le T \\ 0 & \text{otherwise} \end{cases}$$



- Multiplying in the time domain == convolution in the frequency domain
- $F(\mu)$ is band-limited but $H(\mu)$ is not ! (sinc)
- Convolution operation will yield a Fourier Transform extending to infinity in both directions



- Aliasing is almost always present in sampled signals…
- … because we always sample over a limited duration.
- (Exception: band-limited periodic function observed on an exact period)
- (analog) Low-pass filters are always used before sampling !

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Discrete Fourier Transform

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- Let's look for the transform of sampled data starting from the function itself

$$\tilde{F}(\mu) = \int_{-\infty}^{\infty} \tilde{f}(t) e^{-j2\pi\mu t} dt = \int_{-\infty}^{\infty} \sum_{n=-\infty}^{\infty} f(t)\delta(t - n\Delta T) e^{-j2\pi\mu t} dt$$

$$= \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} f(t)\delta(t - n\Delta T) e^{-j2\pi\mu t} dt$$

$$= \sum_{n=-\infty}^{\infty} f_n e^{-j2\pi\mu n\Delta T}$$

- $\tilde{F}(\mu)$ is continuous, periodic of period $1/\Delta T$, so we sample one period by:

$$\mu = \frac{m}{M\Delta T} \qquad m = 0, 1, 2, \ldots, M-1$$

$$\Rightarrow F_m = \sum_{n=0}^{M-1} f_n e^{-j2\pi mn/M} \qquad m = 0, 1, 2, \ldots, M-1$$

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Discrete Fourier Transform

- Discrete Fourier Transform - (forward) DFT:

$$F_m = \sum_{n=0}^{M-1} f_n e^{-j2\pi mn/M} \qquad m = 0, 1, 2, \ldots, M-1$$

- Inverse Discrete Fourier Transform - inverse DFT (IDFT):

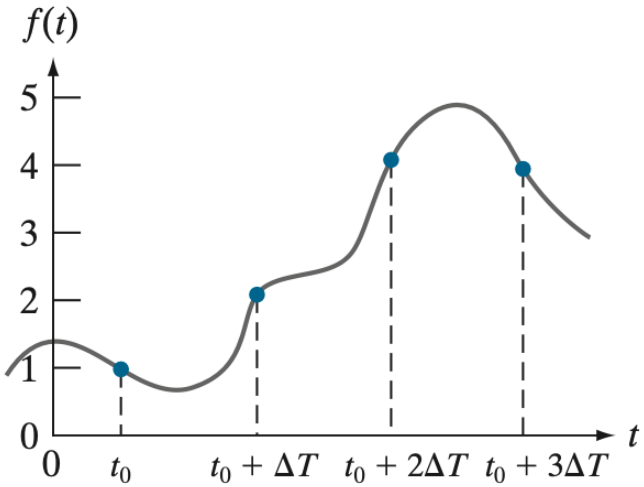
$$f_n = \frac{1}{M} \sum_{m=0}^{M-1} F_m e^{j2\pi mn/M} \quad n = 0, 1, 2, \ldots, M-1$$

- Note: Do not depend on ΔT or frequency interval. Only on uniform sampling.
- With ”image coordinates” notation, we get:

$$F(u) = \sum_{x=0}^{M-1} f(x) e^{-j2\pi ux/M} \qquad u = 0, 1, 2, \ldots, M-1 \qquad f(x) = \frac{1}{M} \sum_{u=0}^{M-1} F(u) e^{j2\pi ux/M} \quad x = 0, 1, 2, \ldots, M-1$$

Discrete Fourier Transform – practical example

- $f(t)$ continuous with 4 samples taken ΔT apart.



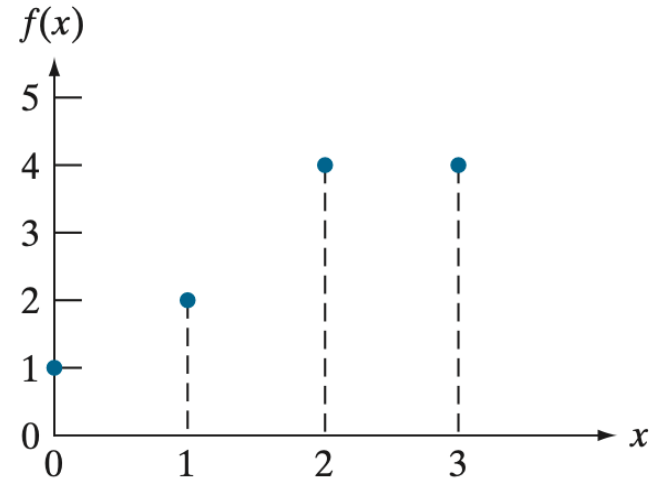
DFT:

$$F(0) = \sum_{x=0}^{3} f(x) = [f(0) + f(1) + f(2) + f(3)] = 1 + 2 + 4 + 4 = 11$$

$$F(1) = \sum_{x=0}^{3} f(x) e^{-j2\pi(1)x/4} = 1e^{0} + 2e^{-j\pi/2} + 4e^{-j\pi} + 4e^{-j3\pi/2} = -3 + 2j$$

$$F(2) = -(1 + 0j) \qquad F(3) = -(3 + 2j)$$

Note: *All* values of $f(x)$ are used in computing *each* value of $F(u)$!



IDFT:

$$f(0) = \frac{1}{4}\sum_{u=0}^{3} F(u) e^{j2\pi u(0)} = \frac{1}{4}\sum_{u=0}^{3} F(u) = \frac{1}{4}[11 - 3 + 2j - 1 - 3 - 2j] = \frac{1}{4}[4] = 1$$

Sampling frequency VS Frequency intervals

- If $f(x)$ has M samples taken ΔT apart, the length of the recording is:

$$T = M\Delta T$$

- The corresponding spacing, Δu, in the frequency domain is then (see previous slides):

$$\Delta u = \frac{1}{M\Delta T} = \frac{1}{T}$$

- The DFT then spans a frequency range:

$$R = M\Delta u = \frac{1}{\Delta T}$$

- The range of frequencies in the DFT depends on sampling ΔT
- The resolution in frequency, Δu, is inversely proportional to the duration of observation

Extension to 2D signals – 2D impulses

- Continuous

$$\delta(t,z)=\begin{cases}\infty & \text{if } t=z=0\\ 0 & \text{otherwise}\end{cases}$$

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\delta(t,z)\,dt\,dz=1$$

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}f(t,z)\delta(t,z)\,dt\,dz=f(0,0)$$

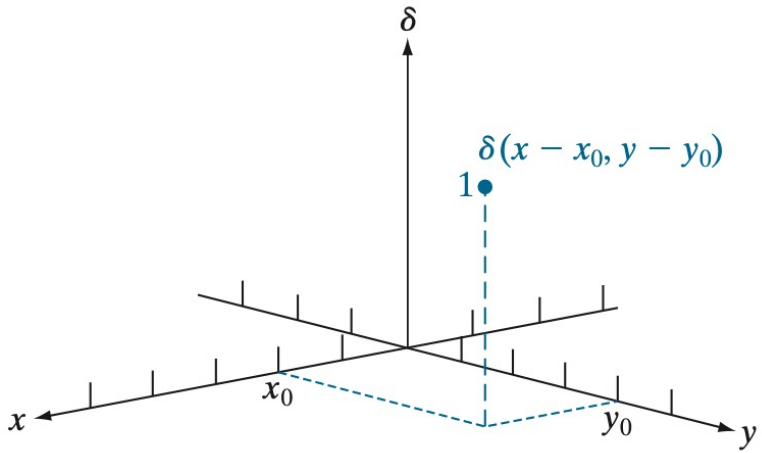
$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}f(t,z)\delta(t-t_0,z-z_0)\,dt\,dz=f(t_0,z_0)$$

- Discrete

$$\delta(x,y)=\begin{cases}1 & \text{if } x=y=0\\ 0 & \text{otherwise}\end{cases}$$

$$\sum_{x=-\infty}^{\infty}\sum_{y=-\infty}^{\infty}f(x,y)\delta(x,y)=f(0,0)$$

$$\sum_{x=-\infty}^{\infty}\sum_{y=-\infty}^{\infty}f(x,y)\delta(x-x_0,y-y_0)=f(x_0,y_0)$$



Extension to 2D signals – 2D Continuous Fourier Transform

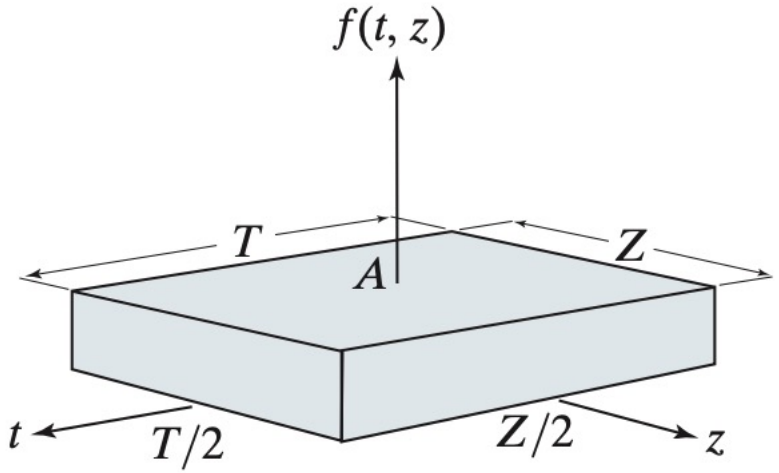
- $f(t,z)$ is a continuous function of 2 continuous variables t and z.

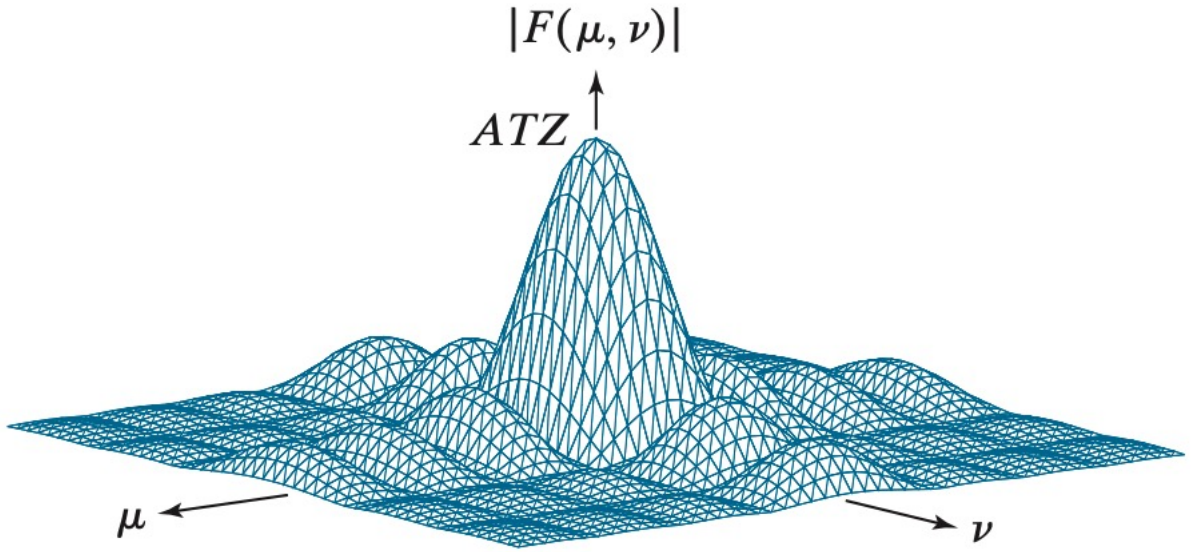
$$F(\mu,\nu) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(t,z)e^{-j2\pi(\mu t+\nu z)}\,dt\,dz \qquad f(t,z) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} F(\mu,\nu)e^{j2\pi(\mu t+\nu z)}\,d\mu\,d\nu$$

where μ and ν are the frequency variables

2D Continuous Fourier Transform – Example

- 2D box-function:





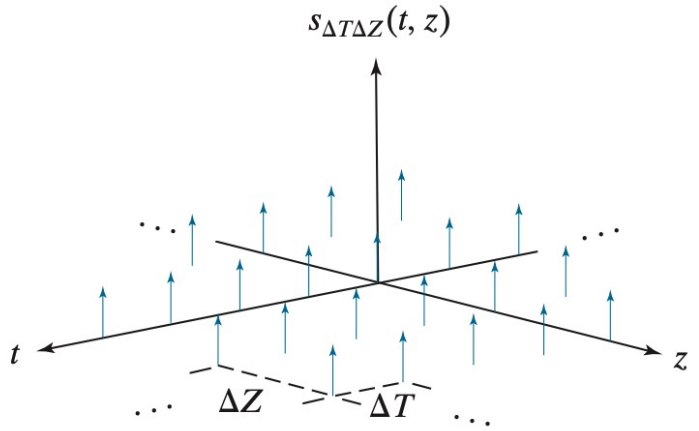
$$F(\mu,\nu) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(t,z)e^{-j2\pi(\mu t+\nu z)}\,dt\,dz = \int_{-T/2}^{T/2}\int_{-Z/2}^{Z/2} A e^{-j2\pi(\mu t+\nu z)}\,dt\,dz$$

$$= ATZ\left[\frac{\sin(\pi\mu T)}{(\pi\mu T)}\right]\left[\frac{\sin(\pi\nu Z)}{(\pi\nu Z)}\right]$$

Extension to 2D signals – 2D Sampling

2D Impulse train:

$$s_{\Delta T \Delta Z}(t,z) = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \delta(t - m\Delta T, z - n\Delta Z)$$

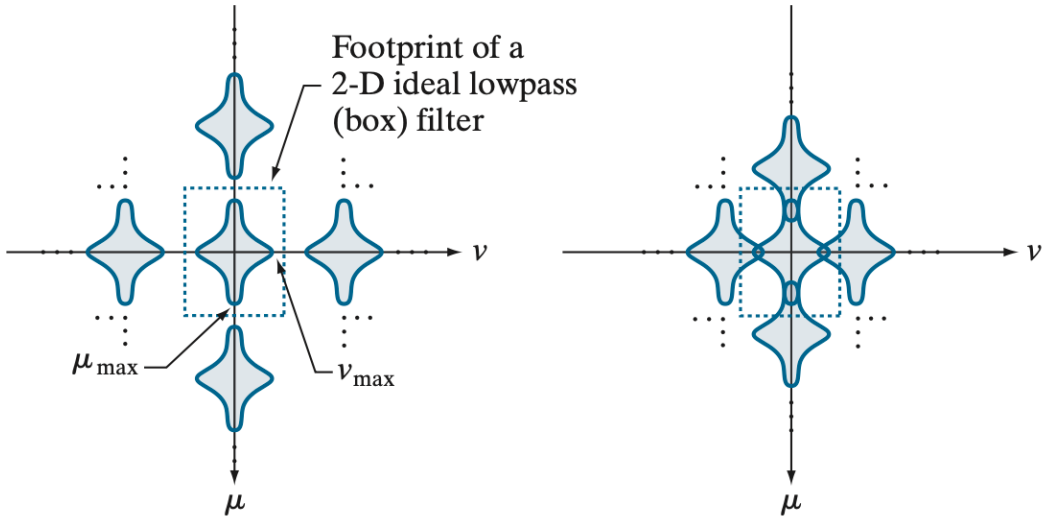


Band limited 2D function:

$$F(\mu,\nu) = 0 \quad \text{for} \quad |\mu| \geq \mu_{\max} \quad \text{and} \quad |\nu| \geq \nu_{\max}$$

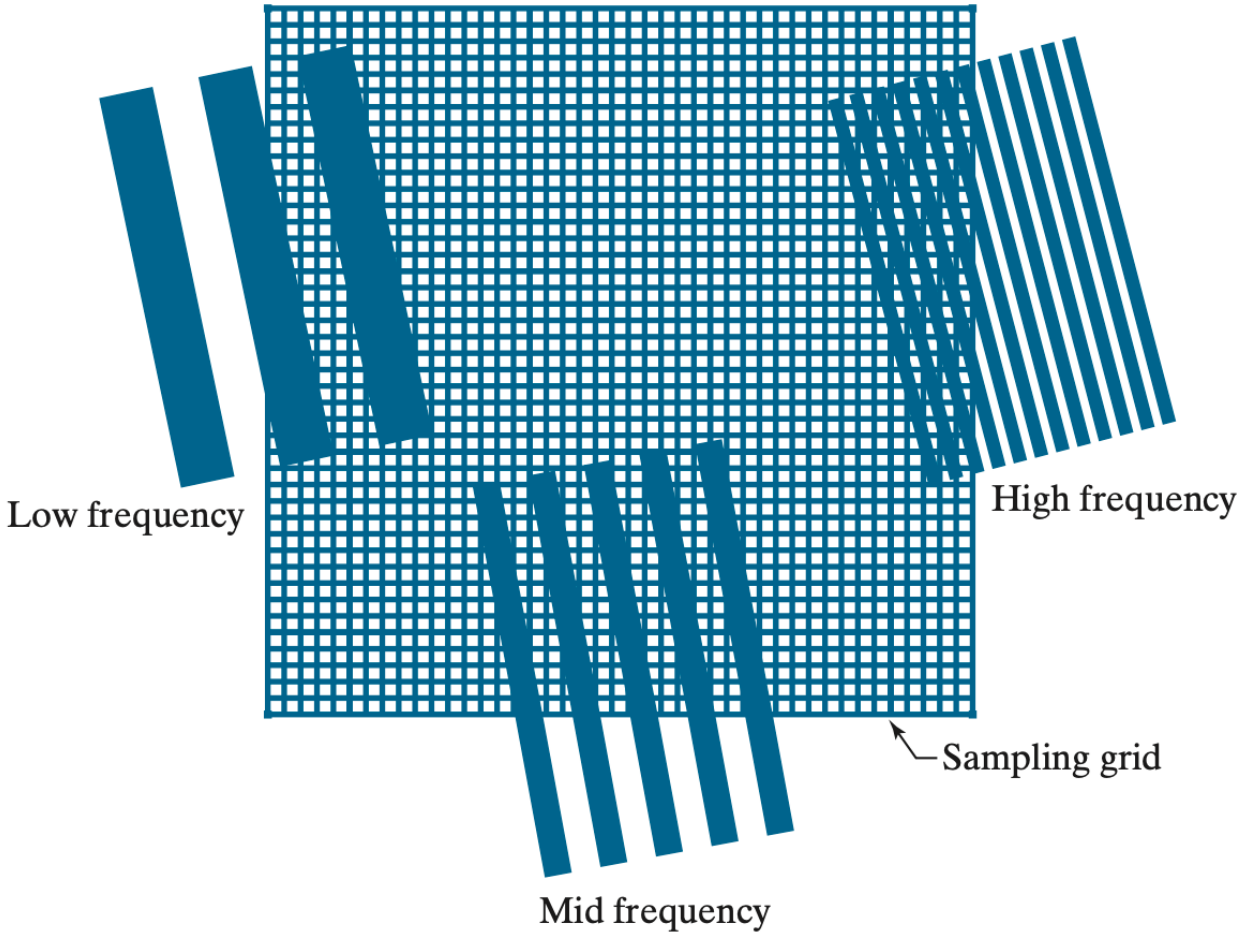
Sampling theorem for a 2D function:

$$\Delta T < \frac{1}{2\mu_{\max}} \qquad \Delta Z < \frac{1}{2\nu_{\max}}$$

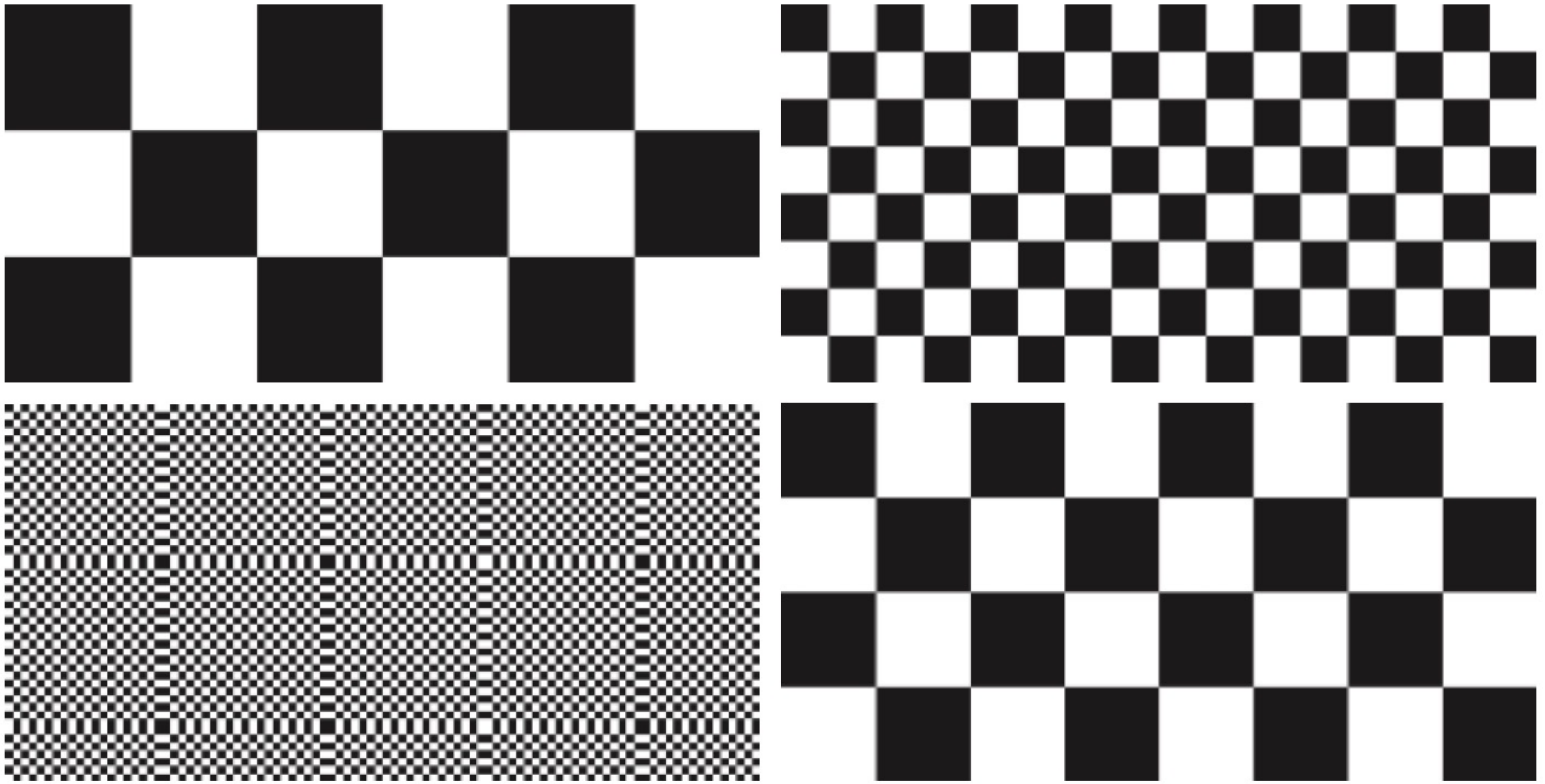


Spatial Aliasing（空间混淆）

- As with 1D-signals, aliasing is always present because images have a finite size !



Spatial Aliasing（空间混淆）



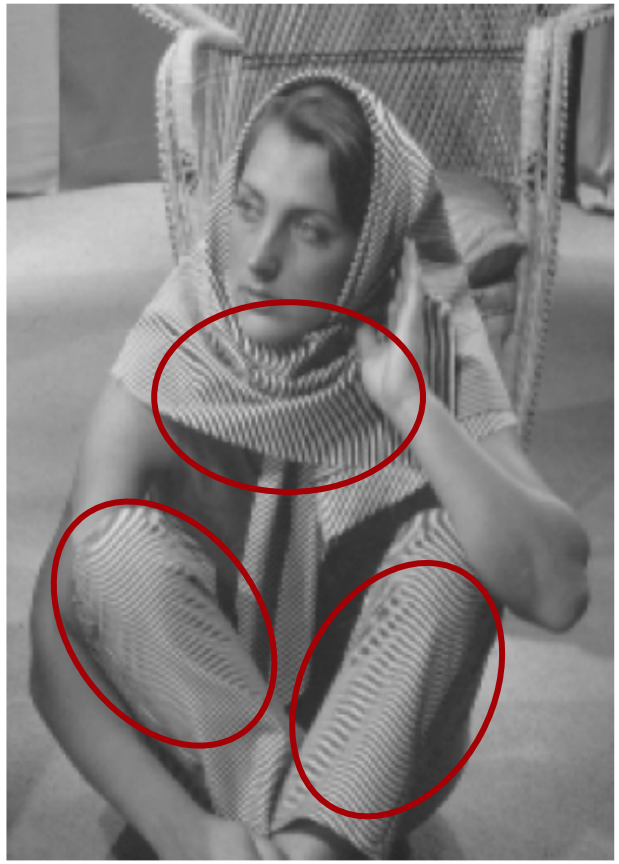
a b
c d

FIGURE 4.18
Aliasing. In (a) and (b) the squares are of sizes 16 and 6 pixels on the side. In (c) and (d) the squares are of sizes 0.95 and 0.48 pixels, respectively. Each small square in (c) is one pixel. Both (c) and (d) are aliased. Note how (d) masquerades as a “normal” image.

Resampling, interpolation and aliasing

- Zooming = over-sampling, shrinking = under-sampling

Original image



Shrunk by 33%

5x5 smoothing filter
Before resizing

2D DFT and IDFT

2D Discrete Fourier Transform (DFT)

$$F(u,v) = \sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi(\frac{ux}{M}+\frac{vy}{N})}$$

2D Inverse Discrete Fourier Transform (IDFT)

$$f(x,y) = \frac{1}{MN}\sum_{u=0}^{M-1}\sum_{v=0}^{N-1} F(u,v)e^{j2\pi(\frac{ux}{M}+\frac{vy}{N})}$$

$$\Delta u = \frac{1}{M\Delta T}$$

$$\Delta v = \frac{1}{N\Delta Z}$$

- $f(x,y)$: M*N input image
- (x,y): spatial variables, $(x = 0,1,2,\cdots,M-1; y = 0,1,2,\cdots,N-1)$
- (u,v): frequency variables, defines the continuous frequency domain $(u = 0,1,2,\cdots,M-1; v = 0,1,2,\cdots,N-1)$.
- **Examples of basis function.**

Basis Functions in 2D DFT

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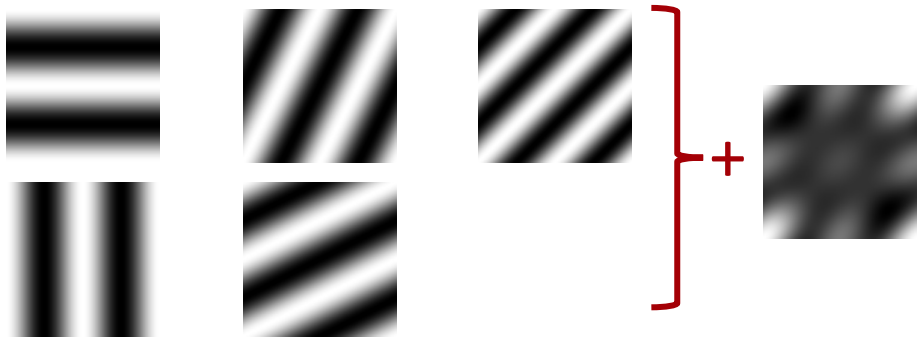
- [0,0]: Constant
- [0,1] [1,0] [1,1]

- [0,2] [2,0] [1,2] [2,1] [2,2]

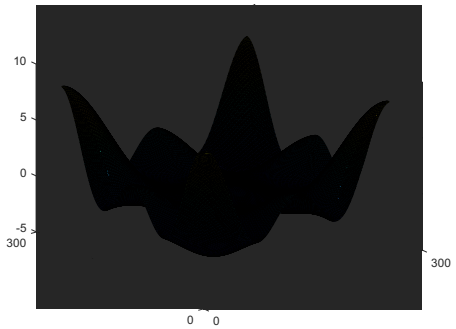


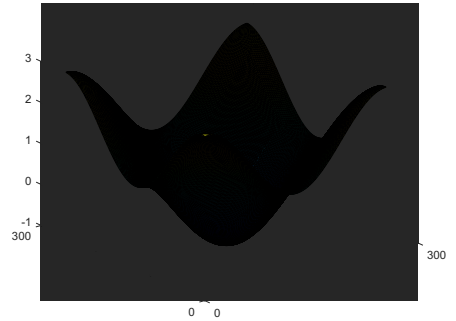
```
function im = bf(m,n)

N = 256;
[x,y] = meshgrid(0:(N-1),0:(N-1));
im = real(exp(-j*2*pi*(m*x/N +n*y/N)));

if (m==0) && (n==0)
    im = round(im);
end

figure; imshow(im,[]);
```





Spectrum and Phase angle

- **2D DFT in polar form:** $F(u,v) = |F(u,v)|e^{-j\Phi(u,v)}$ **, then**
- **Fourier spectrum（频谱）:** $|F(u,v)| = [R^2(u,v) + I^2(u,v)]^{\frac{1}{2}}$
- **Phase angle（相角）:** $\Phi(u,v) = \arctan\frac{I(u,v)}{R(u,v)}$
- **Power spectrum (功率谱):** $P(u,v) = |F(u,v)|^2 = R^2(u,v) + I^2(u,v)$
- **DC component (直流分量):** $F(0,0) = \sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y) = MN\overline{f(x,y)}$

Translation and Rotation

- Translation

$$f(x,y)e^{j2\pi(u_0x/M+v_0y/N)} \Leftrightarrow F(u-u_0,v-v_0)$$

$$f(x-x_0,y-y_0) \Leftrightarrow F(u,v)e^{-j2\pi(x_0u/M+y_0v/N)}$$

  - Translation of the image does not change the spectrum

- We get rotation properties by using polar coordinates

$$x=r\cos\theta \qquad y=r\sin\theta \qquad u=\omega\cos\varphi \qquad v=\omega\sin\varphi$$

$$f(r,\theta+\theta_0) \Leftrightarrow F(\omega,\varphi+\theta_0)$$

  - In case of rotation, image and spectrum are rotated the same way

Periodicity

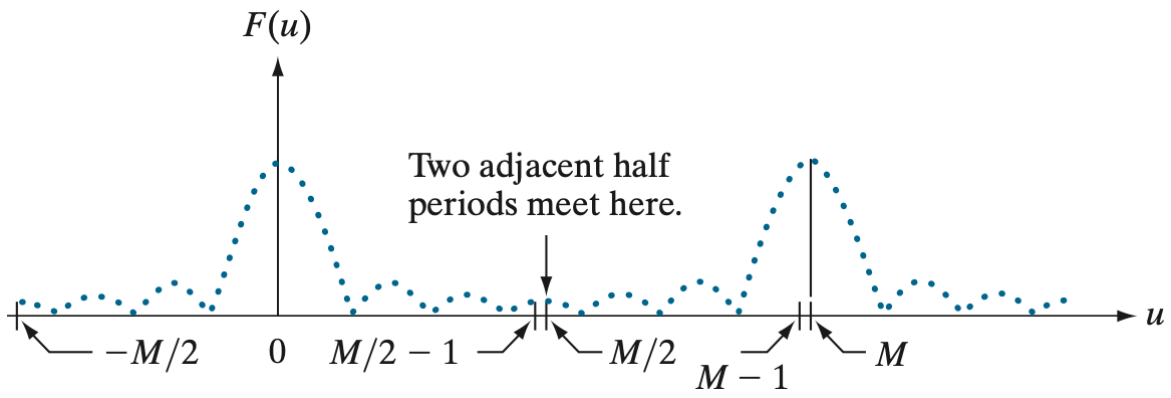
- DFT is periodic

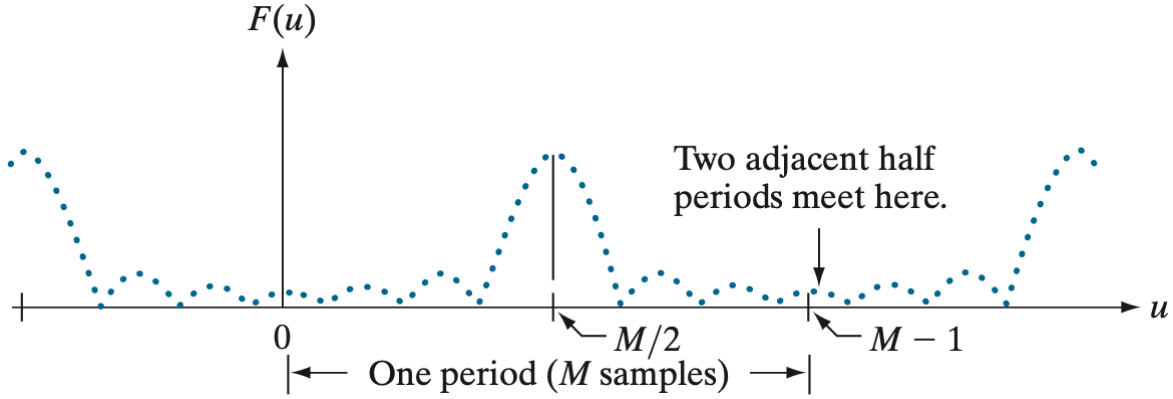
$$F(u,v) = F(u + k_1 M, v) = F(u, v + k_2 N) = F(u + k_1 M, v + k_2 N)$$
$$f(x,y) = f(x + k_1 M, y) = f(x, y + k_2 N) = f(x + k_1 M, y + k_2 N)$$

Where k_1 and k_2 are integers

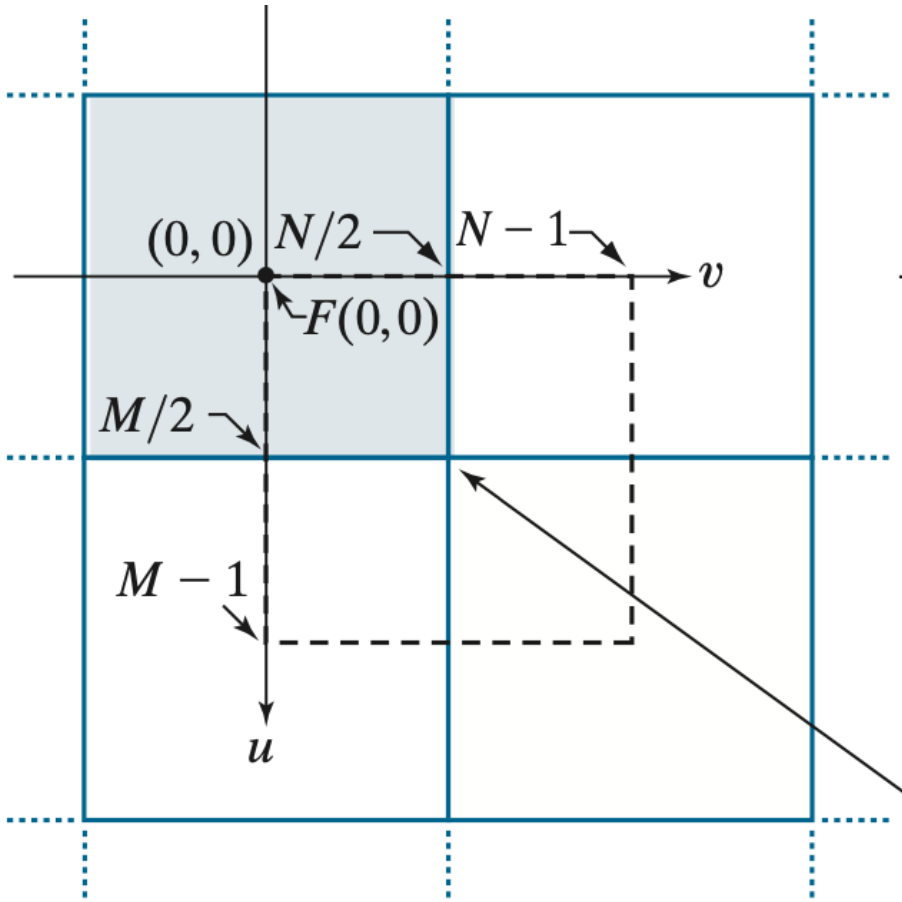
- This can be used to ease representation:

$$f(x)e^{j2\pi(u_0 x/M)} \Leftrightarrow F(u - u_0) \xrightarrow{u_0 = M/2} f(x)(-1)^x \Leftrightarrow F(u - M/2)$$

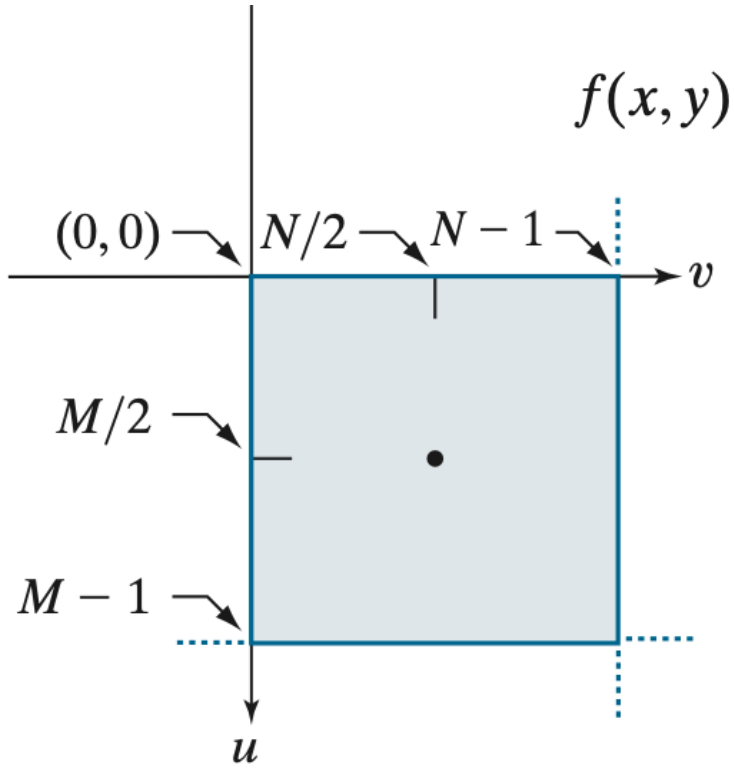




Periodicity



$$f(x,y)(-1)^{x+y} \Leftrightarrow F(u - M/2, v - N/2)$$



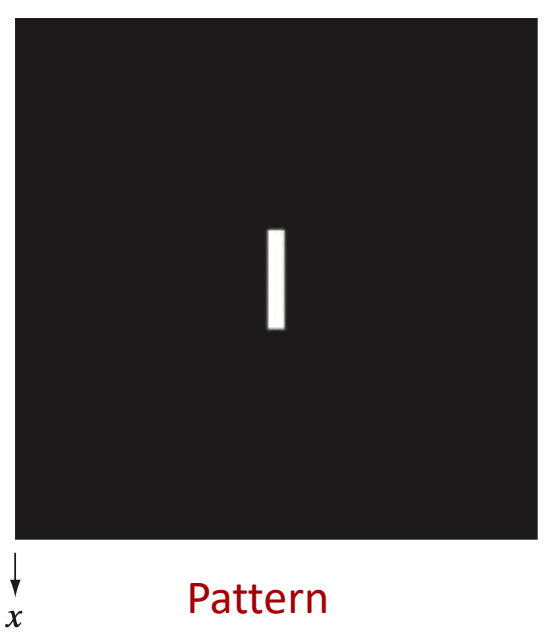
Four adjacent quarter periods meet here

= $M \times N$ data array computed by the DFT with $f(x,y)$ as input

= $M \times N$ data array computed by the DFT with $f(x,y)(-1)^{x+y}$ as input

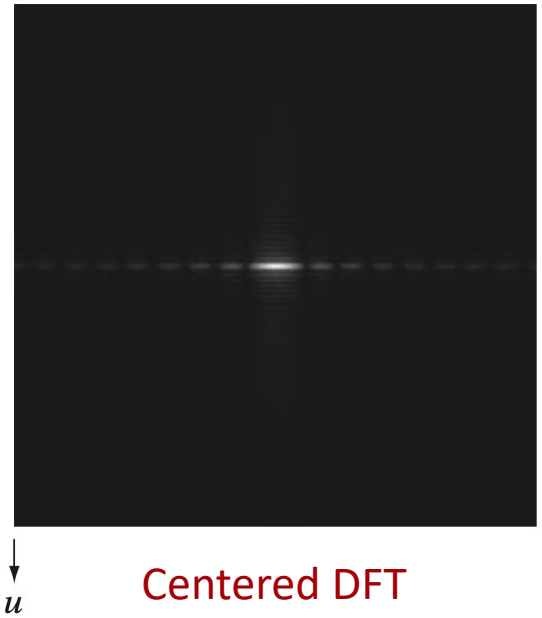
..... = Periods of the DFT

Periodicity

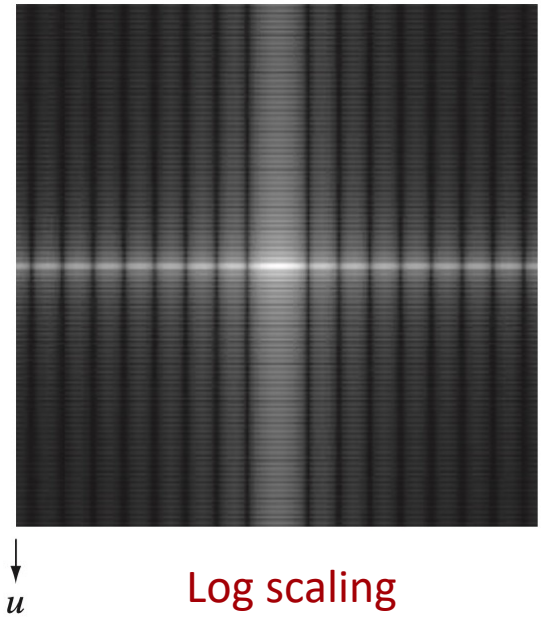


Pattern

DFT

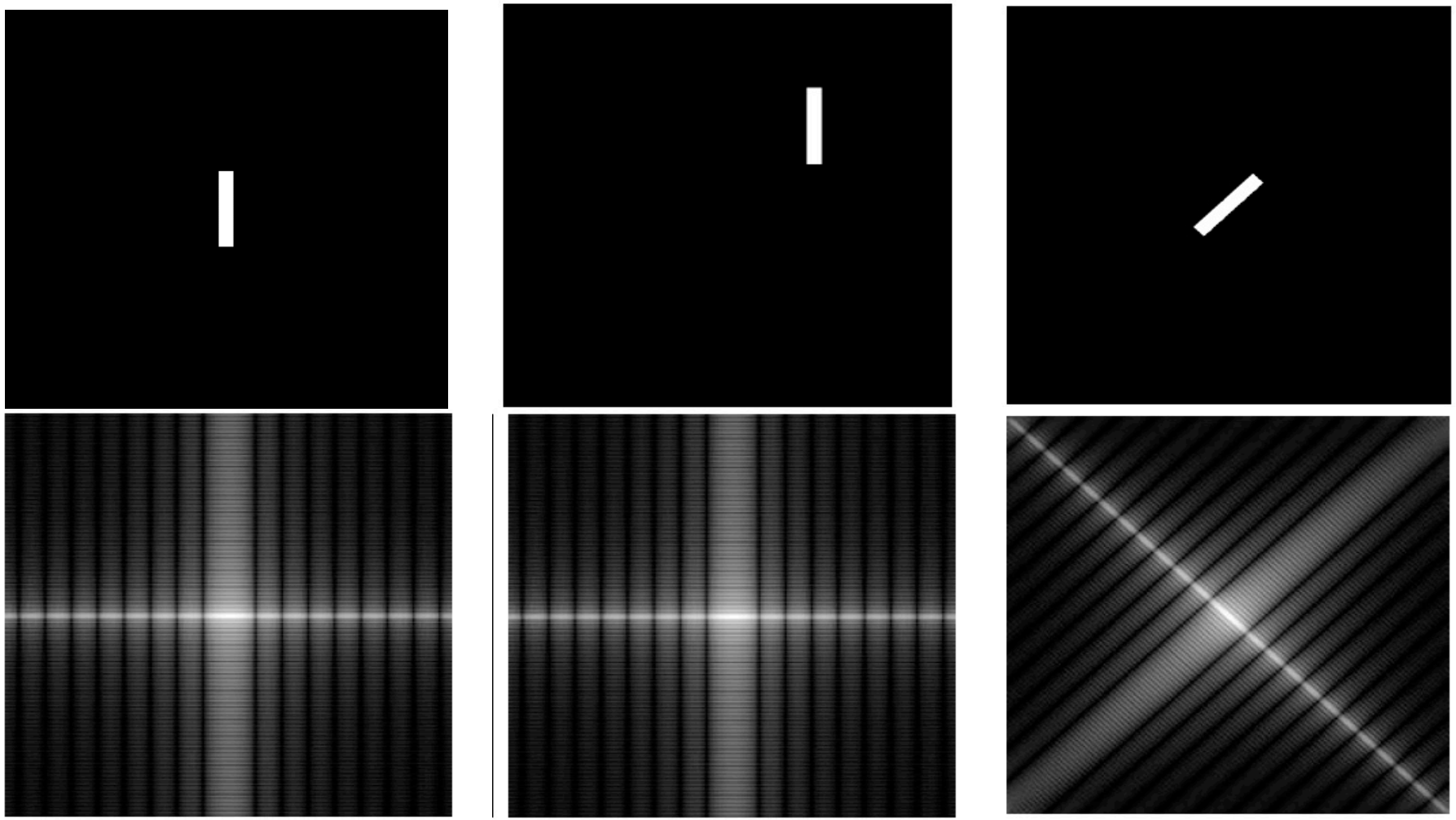


Centered DFT

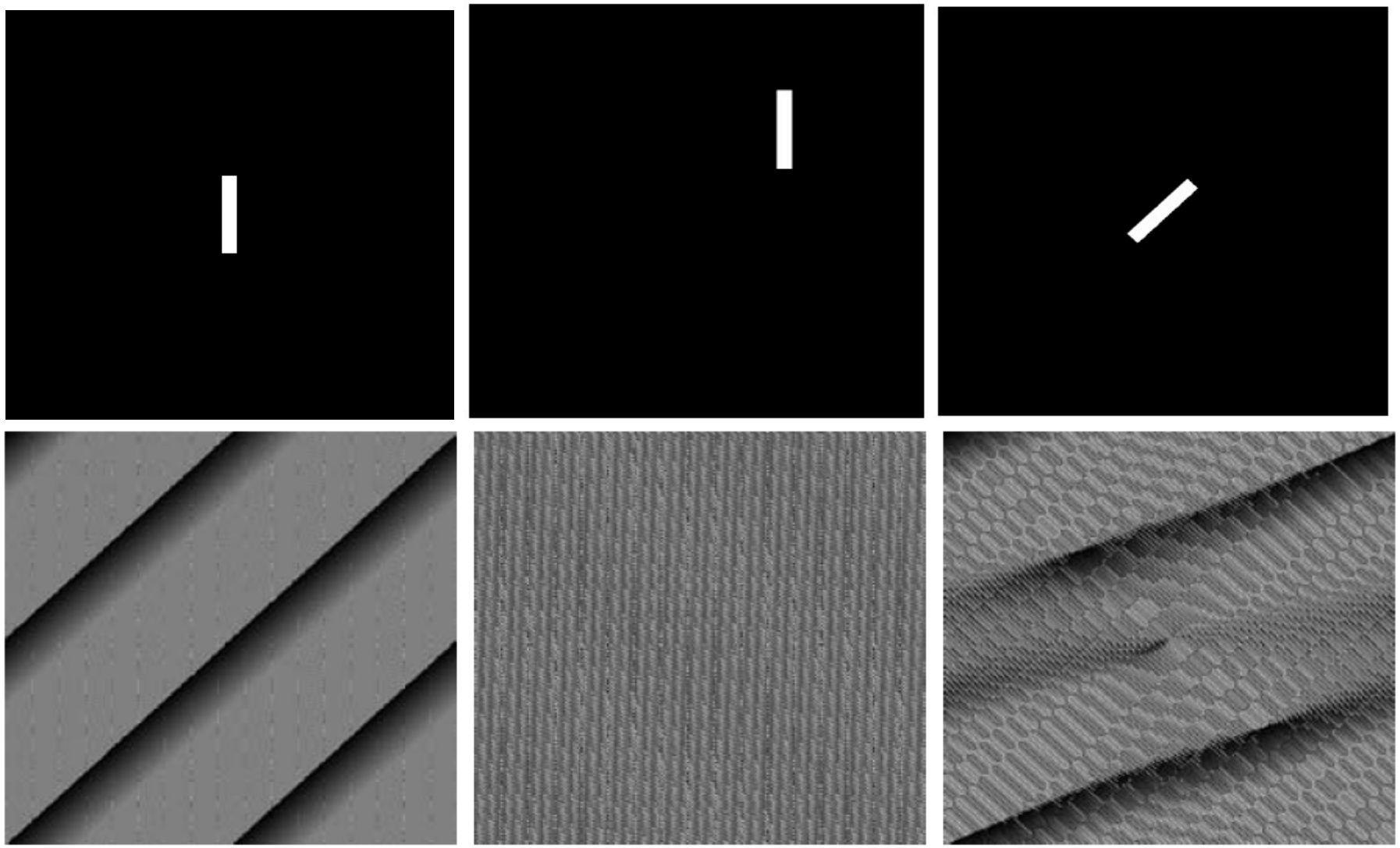


Log scaling

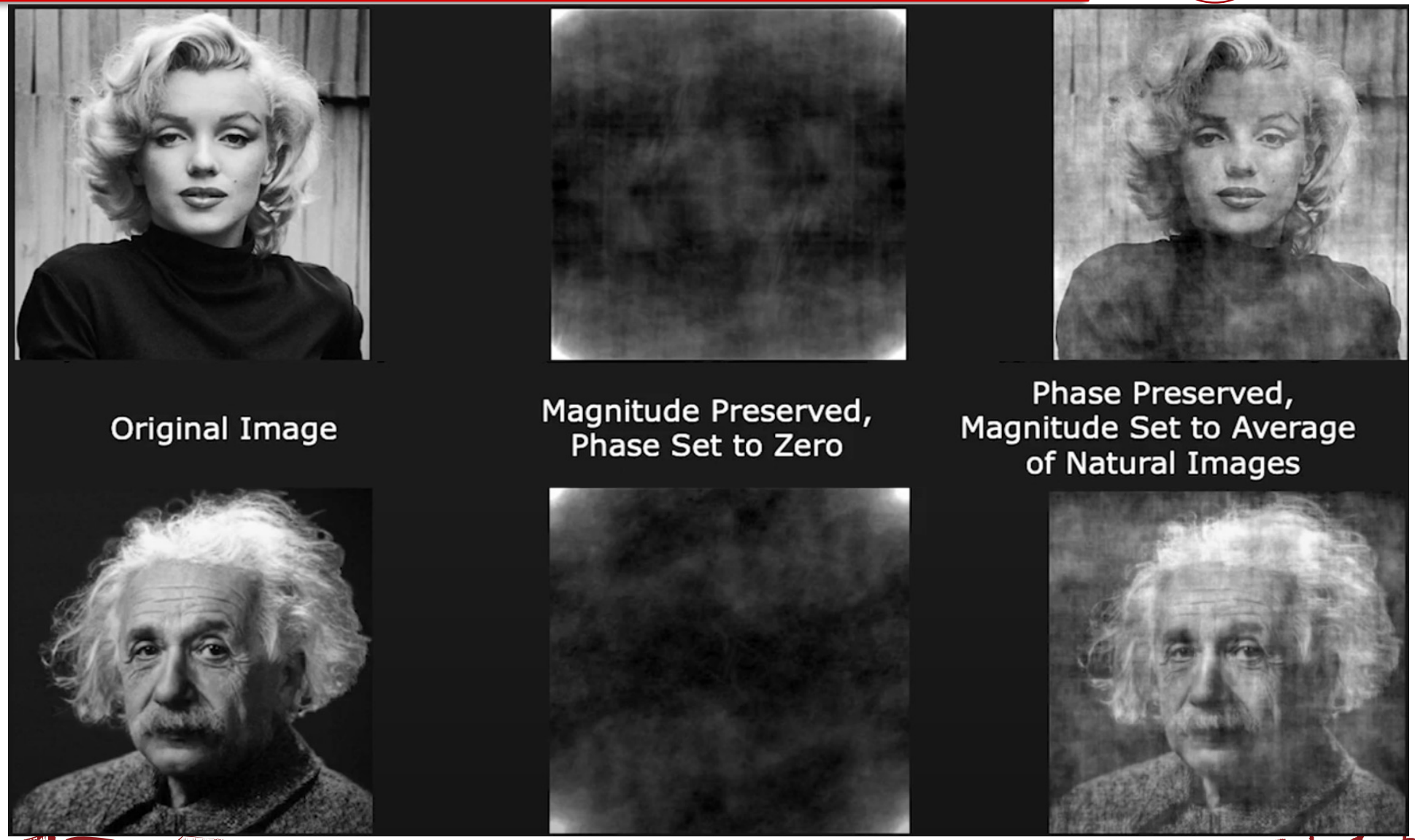
Fourier Spectrum（频谱）



Phase angle（相角）



The Importance of Phase



Spectrum and Phase angle

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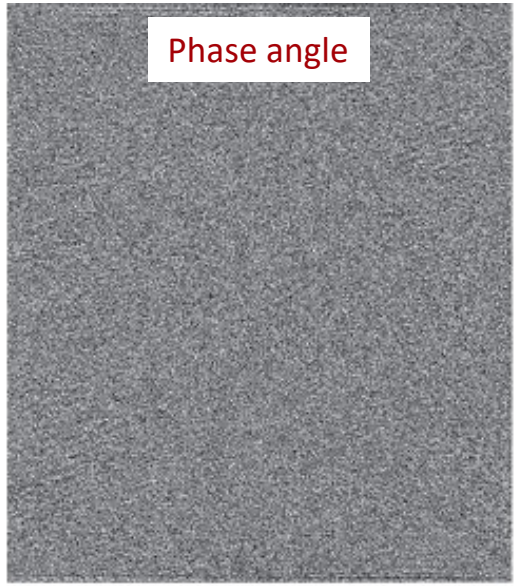
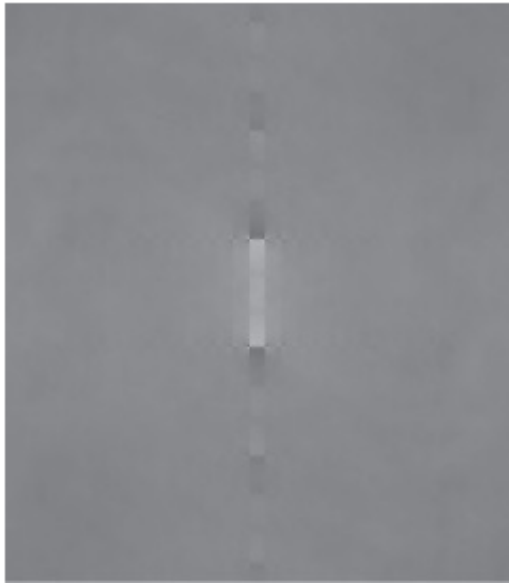
Boy image



Reconstruction
from phase angle only

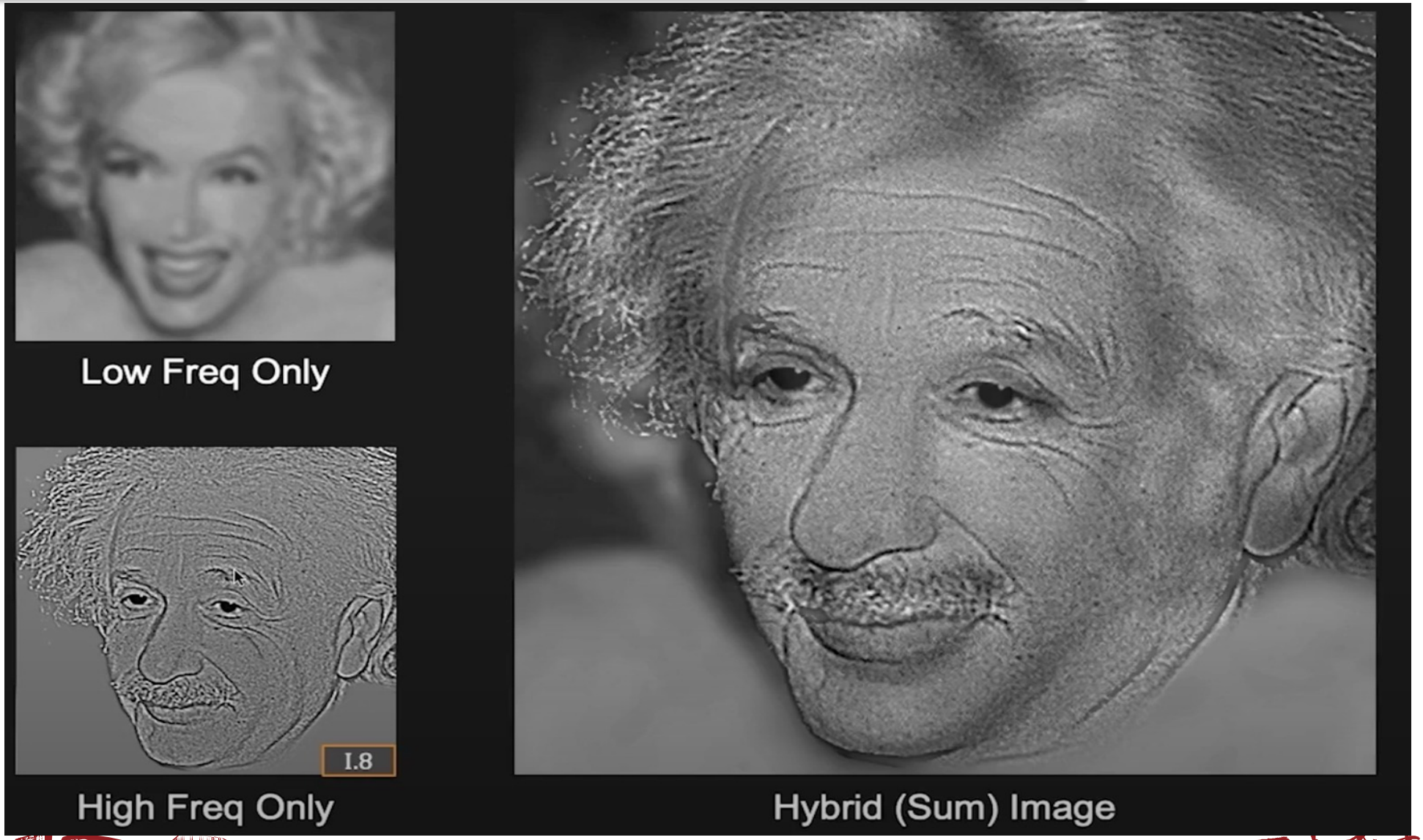
Reconstruction
from spectrum only

Boy angle + rectangle spectrum



Rectangle phase
+ boy spectrum

Monroe or Einstein?



2D Convolution theorem（卷积定理）

- Circular convolution: one period of the periodic convolution

$$(f \star h)(x,y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m,n)h(x-m, y-n)$$

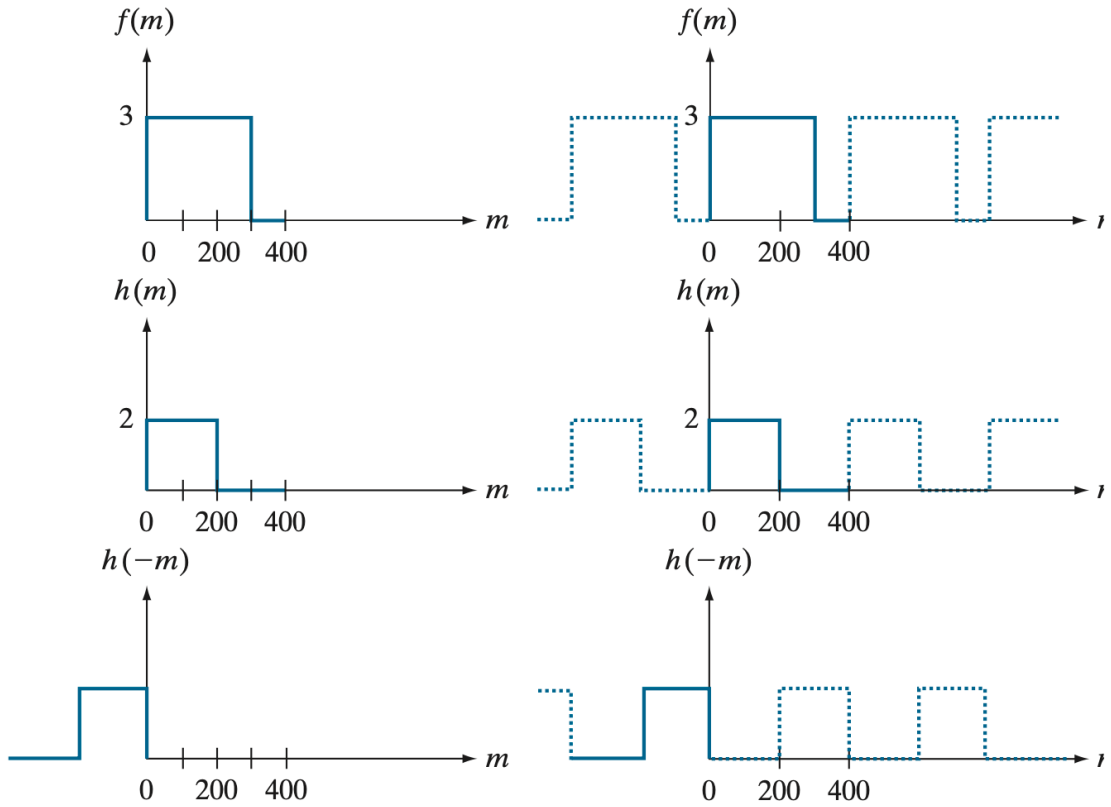
$$\text{for } x = 0, 1, 2, \ldots, M-1 \text{ and } y = 0, 1, 2, \ldots, N-1.$$

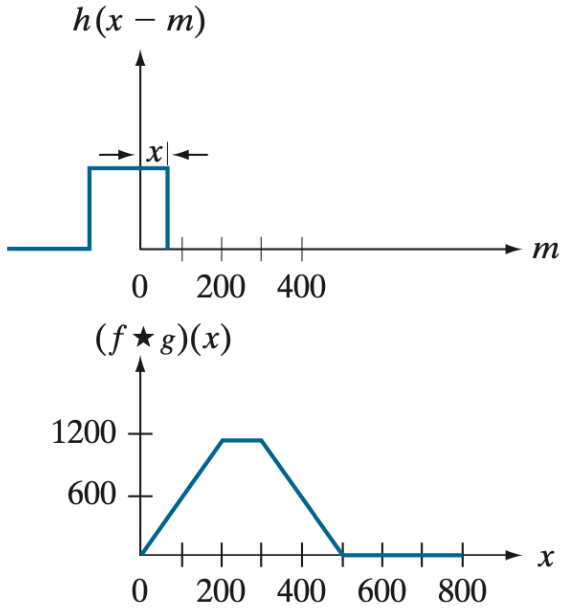
- 2D Convolution theorem:

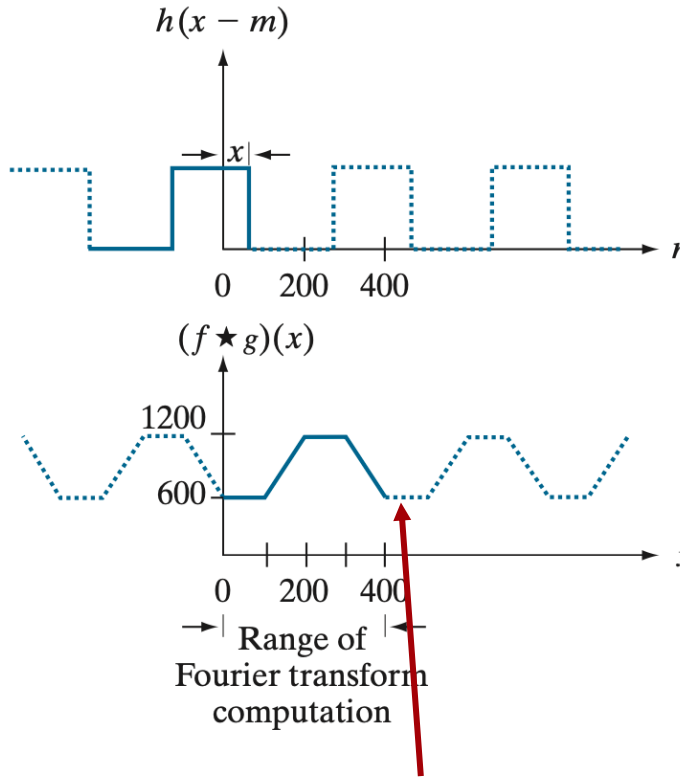
$$(f \star h)(x,y) \Leftrightarrow (F \bullet H)(u,v) \qquad (f \bullet h)(x,y) \Leftrightarrow \frac{1}{MN}(F \star H)(u,v)$$

Circular convolution and DFT

- Implied periodicity !







Solution: Zero-Padding

- Let's consider $f(x)$ has A samples and h(x) has B samples
- We append zeros to both functions so that they have the same length P
- Warparound is avoided by choosing:

$$P \geq A + B - 1$$

- This process is called zero-padding
- This is needed to obtain the same results between "straight" convolution or DFT approach
- In 2D:

$$f_p(x,y) = \begin{cases} f(x,y) & 0 \leq x \leq A-1 \text{ and } 0 \leq y \leq B-1 \\ 0 & A \leq x \leq P \text{ or } B \leq y \leq Q \end{cases}$$

$$h_p(x,y) = \begin{cases} h(x,y) & 0 \leq x \leq C-1 \text{ and } 0 \leq y \leq D-1 \\ 0 & C \leq x \leq P \text{ or } D \leq y \leq Q \end{cases}$$

$$P \geq A + C - 1$$

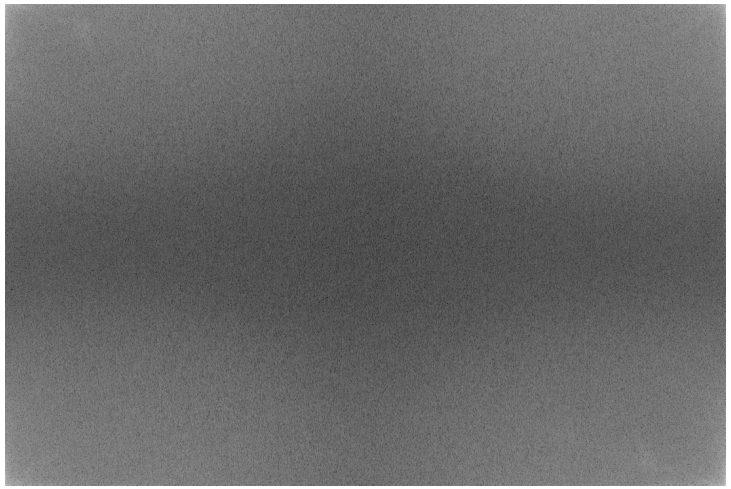
$$Q \geq B + D - 1$$

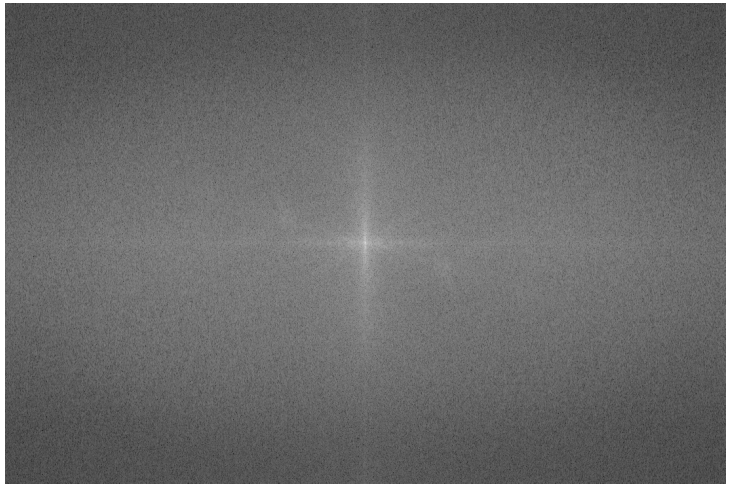
Coding task 1

1. Load image valley-house.png.
2. Compute the DFT
3. Try to show the magnitude of DFT.

$$F(u,v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) e^{-j2\pi(\frac{ux}{M}+\frac{vy}{N})}$$

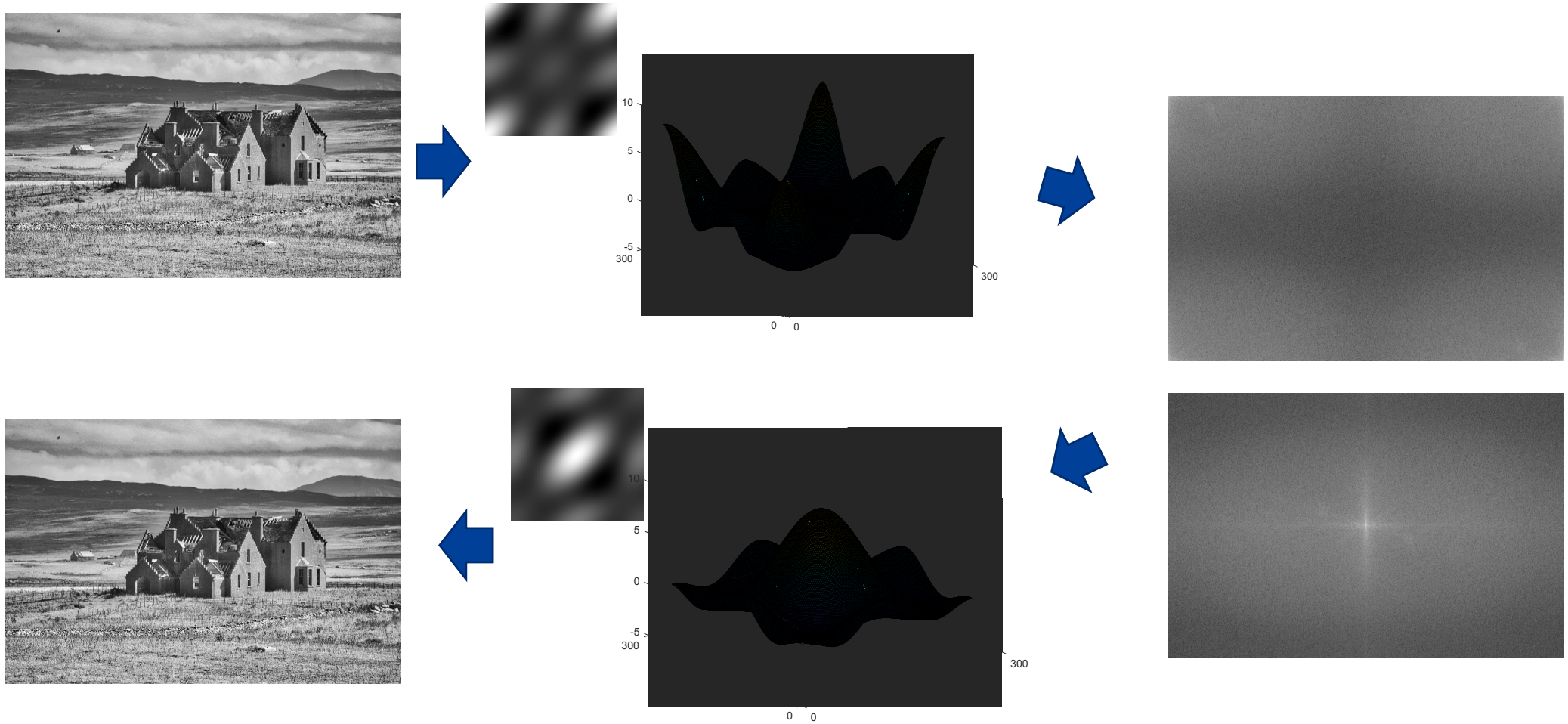
Coding task 1



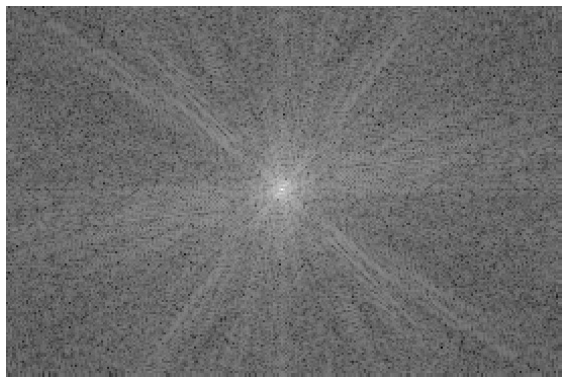


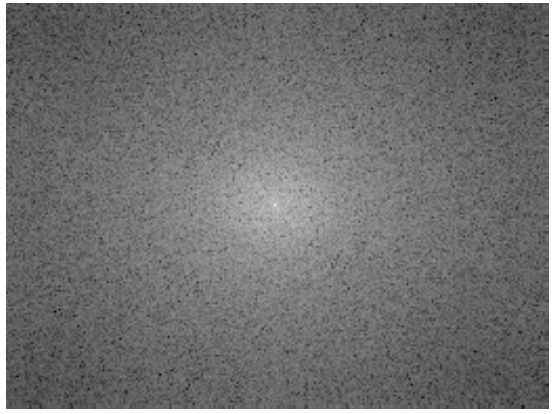
- Load image valley-house.png.
  - fft_im = fft2(im);
- Try to show the magnitude of DFT.
  - fft_im_shifted = fftshift(fft_im);
  - im_recover = ifft2(fft_im);
- Try to show the inverse DFT image.

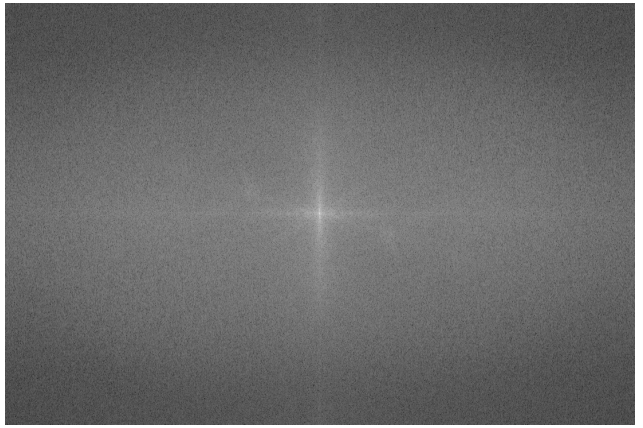
Visualization of DFT



Visualization of DFT







Properties of 2D DFT and IDFT

- Translation （平移）
- Periodicity（周期性）
- Rotation（旋转）
- Separability (可分性)
- Symmetry（对称性）
- 2D Convolution theorem（卷积定理）

Translation（平移）

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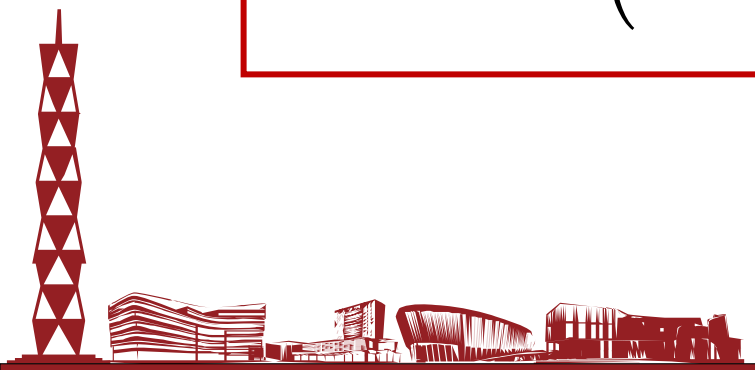
Translation

$$f(x,y)e^{j2\pi(\frac{u_0 x}{M}+\frac{v_0 y}{N})} \Leftrightarrow F(u-u_0, v-v_0)$$

$$f(x-x_0, y-y_0) \Leftrightarrow F(u,v)e^{-j2\pi(\frac{ux_0}{M}+\frac{vy_0}{N})}$$

When $u_0 = \frac{M}{2}, v_0 = \frac{N}{2}$

$$F\left(u-\frac{M}{2}, v-\frac{N}{2}\right) \Leftrightarrow f(x,y)e^{j\pi(x+y)} = f(x,y)(-1)^{(x+y)}$$



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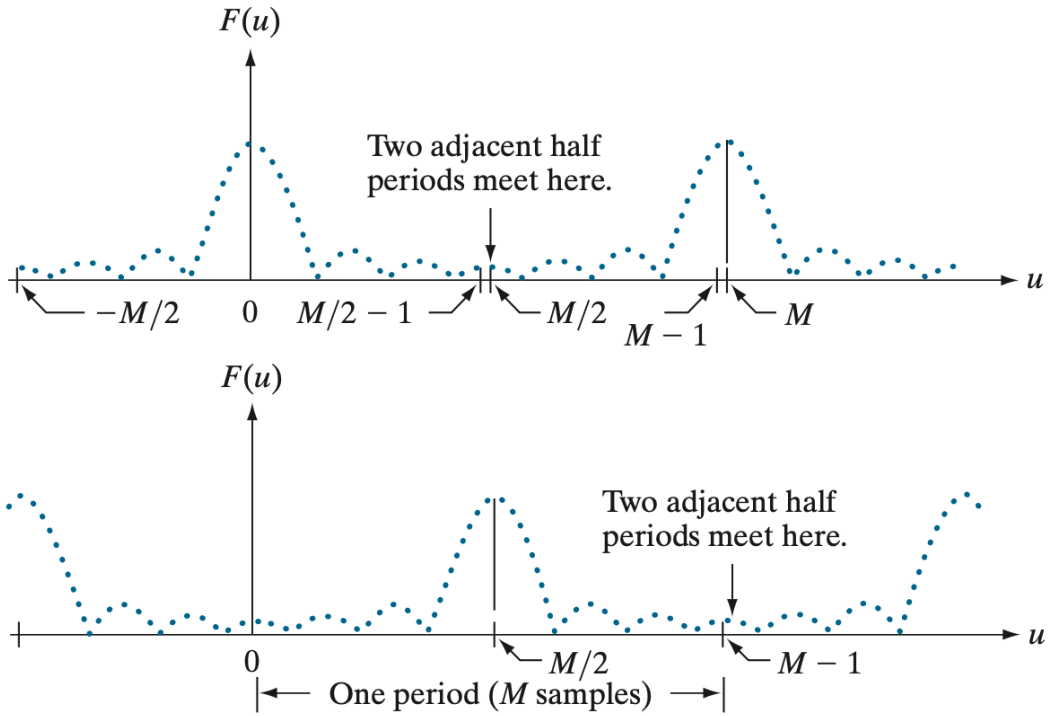
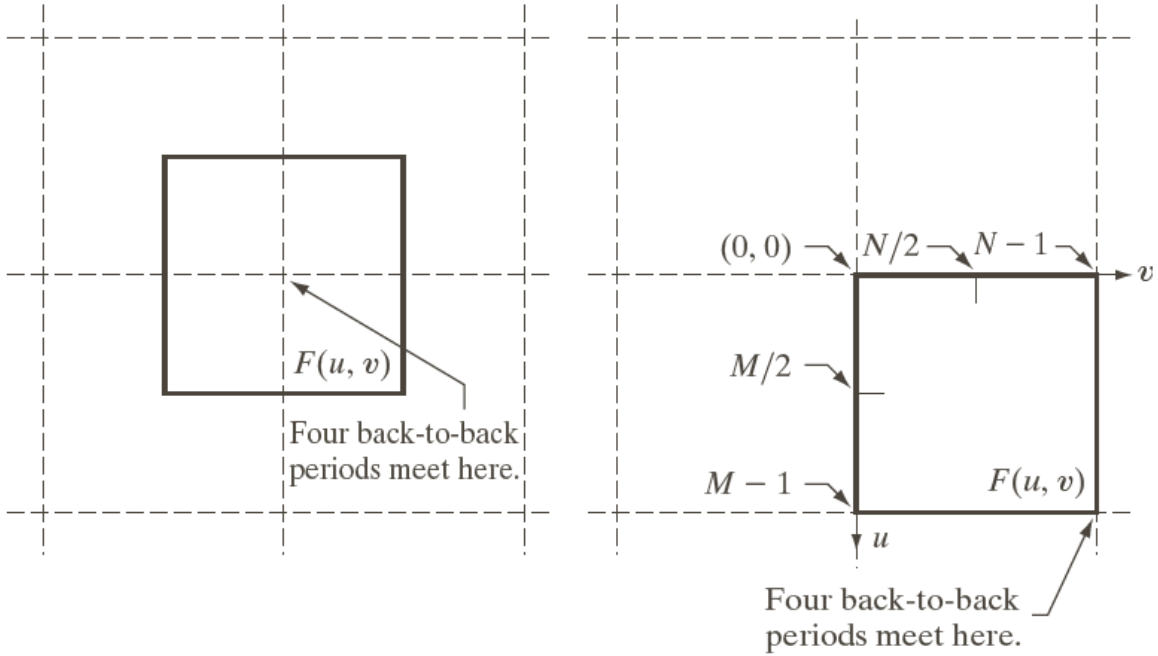
Periodicity（周期性）

- $f(x,y) = f(x + k_1 M, y) = f(x, y + k_2 N) = f(x + k_1 M, y + k_2 N)$
- $F(u,v) = F(u + k_1 M, v) = F(u, v + k_2 N) = F(u + k_1 M, v + k_2 N)$

Where k_1 and k_2 are integers



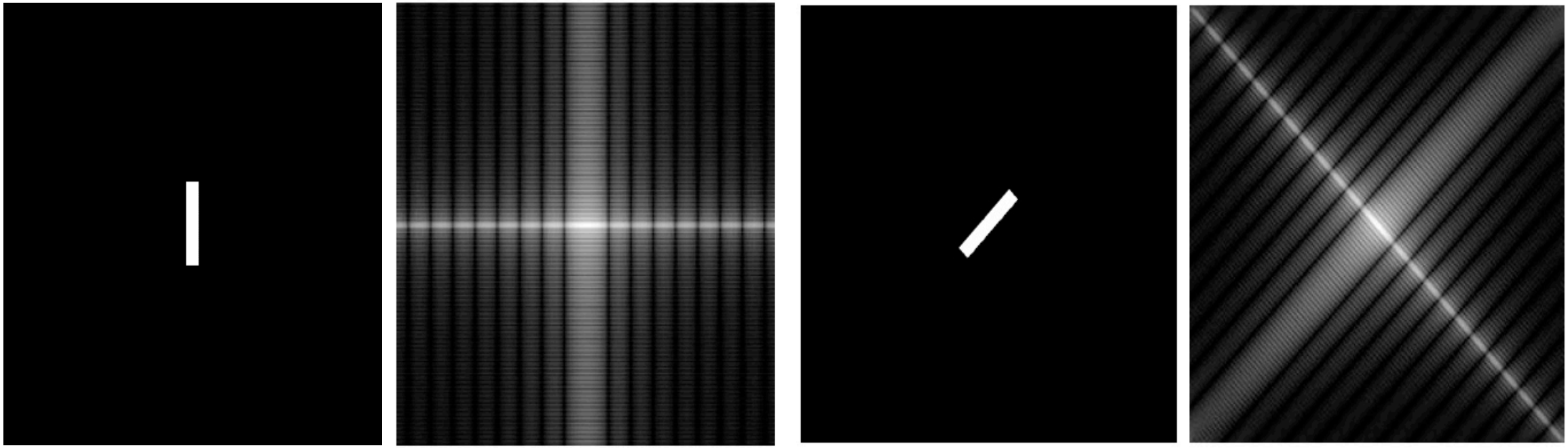


Rotation（旋转）

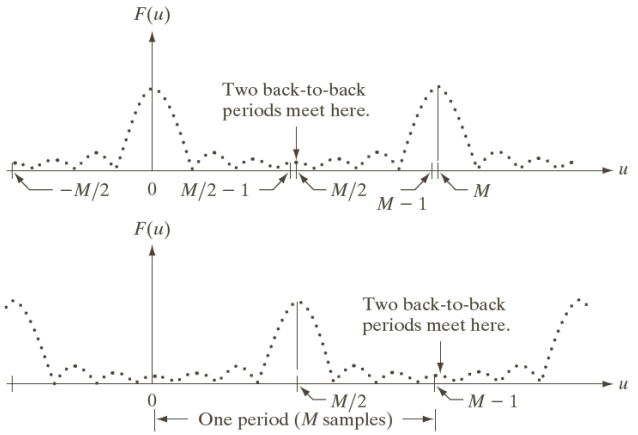
- Rotation

$$f(r, \theta + \theta_0) \Leftrightarrow F(\omega, \varphi + \theta_0)$$

Where $x = r\cos\theta$, $y = r\sin\theta$, $u = \omega\cos\varphi$, $v = \omega\sin\varphi$



Symmetry（对称性）



- Even Function (偶函数)

$$w_e(x,y) = w_e(-x,-y) \quad w_e(x,y) = w_e(M-x,N-y)$$

- Odd Function (奇函数)

$$w_o(x,y) = -w_o(-x,-y) \quad w_o(x,y) = -w_o(M-x,N-y)$$

- Conjugate symmetric (共轭对称) if $f(x,y)$ is real

$$F^*(u,v) = F(-u,-v) \quad F^*(u,v) = F(M-u,N-v)$$

- Conjugate antisymmetric (共轭反对称) if $f(x,y)$ is imaginary

$$F^*(u,v) = -F(-u,-v) \quad F^*(u,v) = -F(M-u,N-v)$$

Symmetry（对称性）

	Spatial Domain†		Frequency Domain†
1)	$f(x, y)$ real	$\Leftrightarrow$	$F^*(u, v) = F(-u, -v)$
2)	$f(x, y)$ imaginary	$\Leftrightarrow$	$F^*(-u, -v) = -F(u, v)$
3)	$f(x, y)$ real	$\Leftrightarrow$	$R(u, v)$ even; $I(u, v)$ odd
4)	$f(x, y)$ imaginary	$\Leftrightarrow$	$R(u, v)$ odd; $I(u, v)$ even
5)	$f(-x, -y)$ real	$\Leftrightarrow$	$F^*(u, v)$ complex
6)	$f(-x, -y)$ complex	$\Leftrightarrow$	$F(-u, -v)$ complex
7)	$f^*(x, y)$ complex	$\Leftrightarrow$	$F^*(-u - v)$ complex
8)	$f(x, y)$ real and even	$\Leftrightarrow$	$F(u, v)$ real and even
9)	$f(x, y)$ real and odd	$\Leftrightarrow$	$F(u, v)$ imaginary and odd
10)	$f(x, y)$ imaginary and even	$\Leftrightarrow$	$F(u, v)$ imaginary and even
11)	$f(x, y)$ imaginary and odd	$\Leftrightarrow$	$F(u, v)$ real and odd
12)	$f(x, y)$ complex and even	$\Leftrightarrow$	$F(u, v)$ complex and even
13)	$f(x, y)$ complex and odd	$\Leftrightarrow$	$F(u, v)$ complex and odd

Accelerating DFT and IDFT computations

- As we have seen, DFT and IDFT are fundamental to frequency domain filtering…
- … but they also require a lot of computation

- In the following slides we will derive some methods to optimize this process leading to…

The Fast Fourier Transform (FFT)

Separability

$$F(u,v) = \sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi(ux/M + vy/N)}$$

$$F(u,v) = \sum_{x=0}^{M-1} e^{-j2\pi ux/M} \sum_{y=0}^{N-1} f(x,y)e^{-j2\pi vy/N}$$

$$= \sum_{x=0}^{M-1} F(x,v)e^{-j2\pi ux/M}$$

where $F(x,v) = \sum_{y=0}^{N-1} f(x,y)e^{-j2\pi vy/N}$

1D DFT of one row of $f(x,y)$

- 2D DFT of $f(x,y)$ ==
  - First: 1D Transform of each *row* of $f(x,y)$
  - Then: 1D Transform along each *column* of the result
- Or 2D DFT of $f(x,y)$ ==
  - First: 1D Transform of each *column* of $f(x,y)$
  - Then: 1D Transform along each *row* of the result

- We could use the same approach for the IDFT but we prefer to use a DFT algorithm for it too

Computing the IDFT with a DFT algorithm

$$f(x,y) = \frac{1}{MN}\sum_{u=0}^{M-1}\sum_{v=0}^{N-1} F(u,v)e^{j2\pi(ux/M + vy/N)}$$

$$MNf^*(x,y) = \sum_{u=0}^{M-1}\sum_{v=0}^{N-1} F^*(u,v)e^{-j2\pi(ux/M + vy/N)}$$

DFT of $F^*(u,v)$

- Input: $F(u,v)$

1. Take the complex conjugate → $F^*(u,v)$
2. Apply forward DFT algorithm → $MNf^*(x,y)$
3. Take the complex conjugate → $MNf(x,y)$
4. Divide by MN → $f(x,y)$

- An optimized forward DFT is all we need !

2D DFT implementation (FFT)

- **Fast Fourier transform (FFT):** an algorithm that computes the discrete Fourier transform (DFT) of a 1D sequence.
- The basic idea is to **decompose** successively the N-point DFT into smaller-size DFTs.
- Improvement of speed $\boldsymbol{O(N^2)}$ to $\boldsymbol{O(Nlog_2(N))}$
- There are many FFT algorithms!
- A classical one is Cooley-Tukey FFT (published in 1965; for more generalized complex FFT), the simplest form is also referred to the *successive-doubling method*

Cooley-Tukey FFT - Successive-Doubling Method

- We express the FFT as:

$$F(u) = \sum_{x=0}^{M-1} f(x) W_M^{ux} \qquad W_M = e^{-j2\pi/M}$$

- And we assume:

$$M = 2^p \qquad M = 2K$$

- Substituting gives us

$$F(u) = \sum_{x=0}^{2K-1} f(x) W_{2K}^{ux}$$

$$= \sum_{x=0}^{K-1} f(2x) W_{2K}^{u(2x)} + \sum_{x=0}^{K-1} f(2x+1) W_{2K}^{u(2x+1)}$$

- And we can show that

$$W_{2K}^{2ux} = W_K^{ux}$$

Cooley-Tukey FFT - Successive-Doubling Method

- Which gives us:

$$F(u) = \underbrace{\sum_{x=0}^{K-1} f(2x) W_K^{ux}}_{F_{even}(u)} + \underbrace{\sum_{x=0}^{K-1} f(2x+1) W_K^{ux}}_{F_{odd}(u)} W_{2K}^{u}$$

$$F(u) = F_{\text{even}}(u) + F_{\text{odd}}(u) W_{2K}^{u}$$

- We can also show that:

$$W_K^{u+K} = W_K^u \; and \; W_{2K}^{u+K} = -W_{2K}^u$$

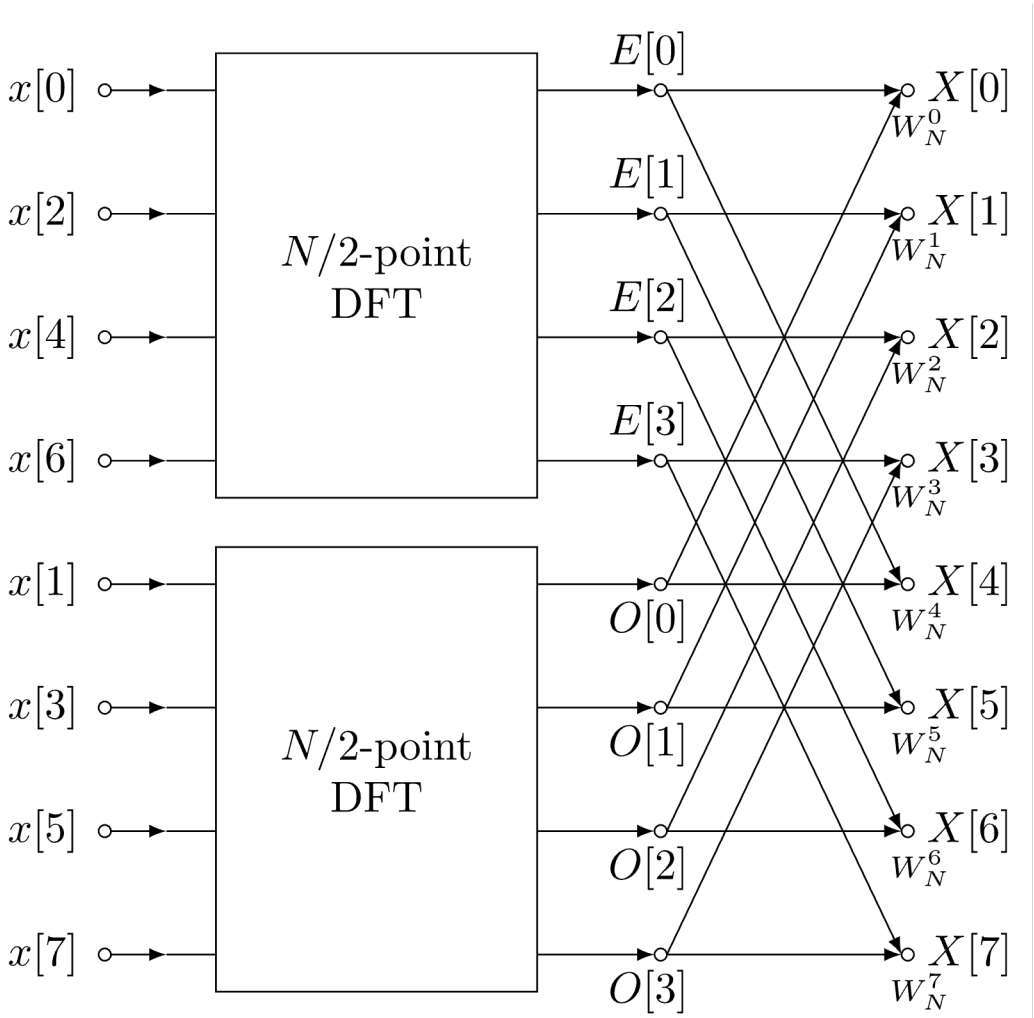
- Substituting gives us

$$F(u+K) = F_{\text{even}}(u) - F_{\text{odd}}(u) W_{2K}^{u}$$

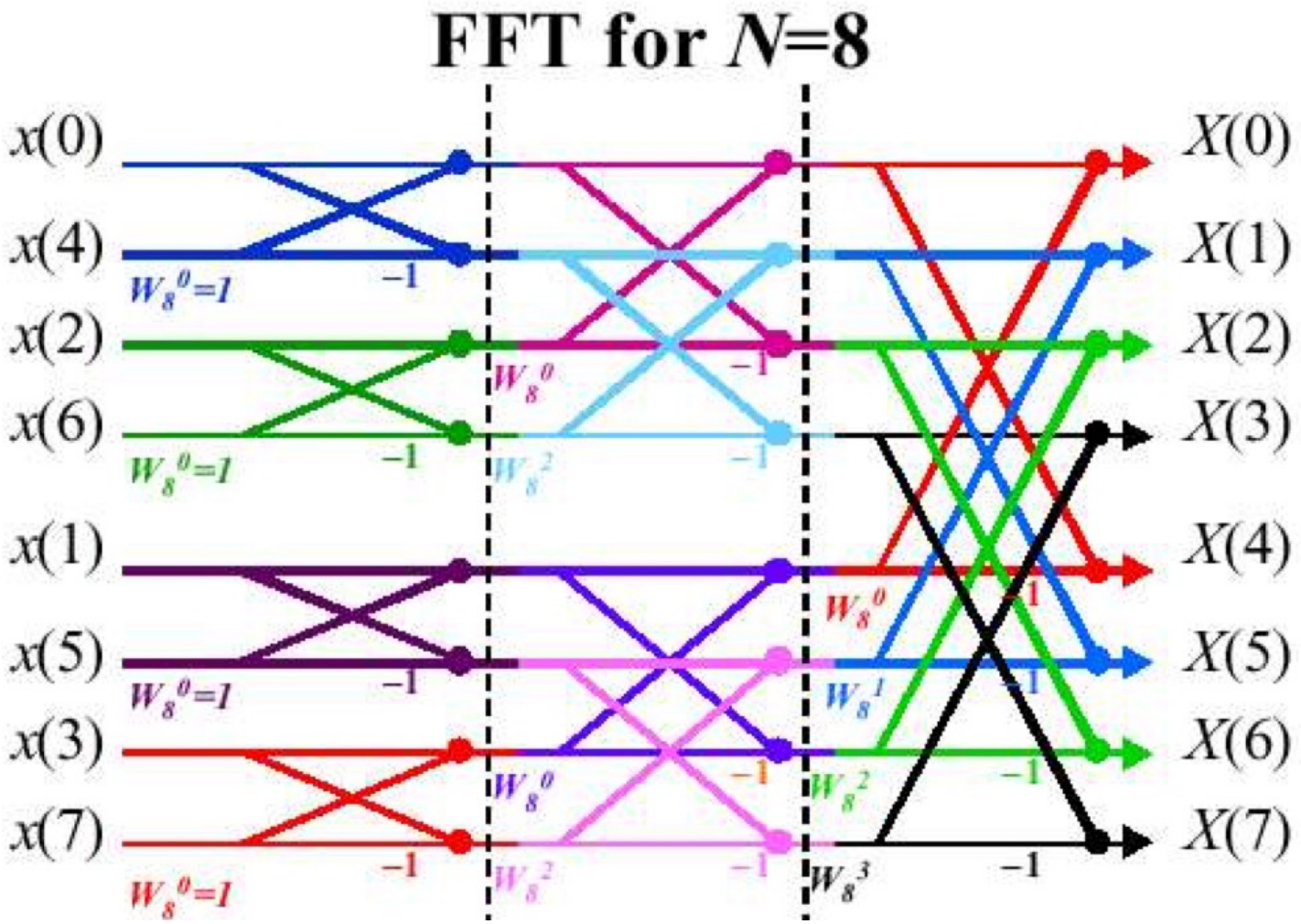
- So we need to compute 2 half DFTs that can still be decomposed !

Cooley-Tukey FFT - Successive-Doubling Method

$$
\begin{aligned}
X_k &= E_k + e^{-\frac{2\pi i}{N}k} O_k \\
X_{k+\frac{N}{2}} &= E_k - e^{-\frac{2\pi i}{N}k} O_k
\end{aligned}
$$



Cooley-Tukey FFT - Successive-Doubling Method



Cooley-Tukey FFT - Successive-Doubling Method

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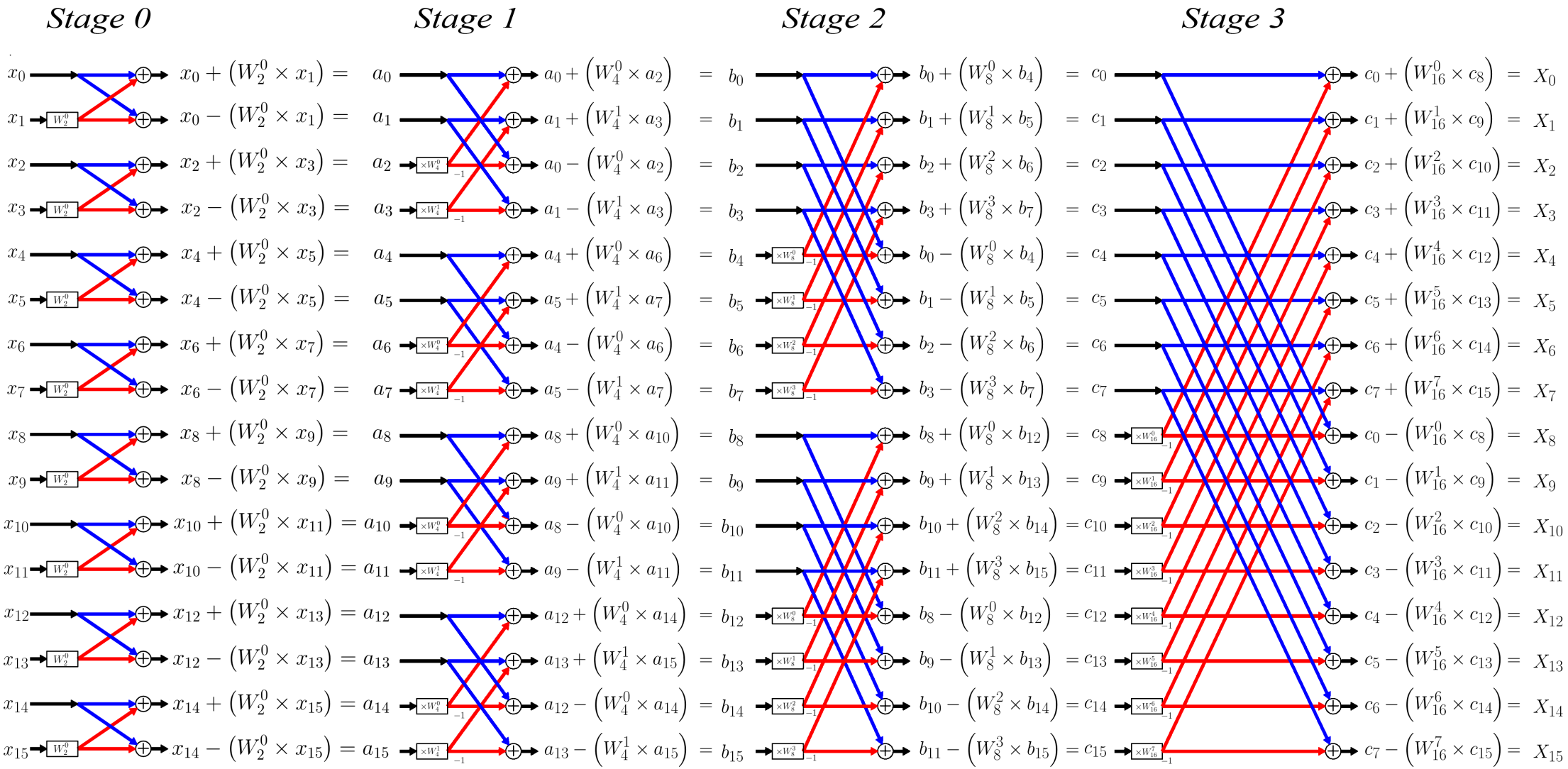

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Appendix 1: DFT definition and expressions

TABLE 4.3
Summary of DFT definitions and corresponding expressions.

	Name	Expression(s)
1)	Discrete Fourier transform (DFT) of $f(x,y)$	$F(u,v) = \sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi(ux/M+vy/N)}$
2)	Inverse discrete Fourier transform (IDFT) of $F(u,v)$	$f(x,y) = \frac{1}{MN}\sum_{u=0}^{M-1}\sum_{v=0}^{N-1} F(u,v)e^{j2\pi(ux/M+vy/N)}$
3)	Spectrum	$\lvert F(u,v)\rvert = \left[R^2(u,v) + I^2(u,v)\right]^{1/2}$ $R = \text{Real}(F)$; $I = \text{Imag}(F)$
4)	Phase angle	$\phi(u,v) = \tan^{-1}\left[\frac{I(u,v)}{R(u,v)}\right]$
5)	Polar representation	$F(u,v) = \lvert F(u,v)\rvert e^{j\phi(u,v)}$
6)	Power spectrum	$P(u,v) = \lvert F(u,v)\rvert^2$
7)	Average value	$\bar{f} = \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y) = \frac{1}{MN}F(0,0)$
8)	Periodicity (k_1 and k_2 are integers)	$F(u,v) = F(u+k_1M,v) = F(u,v+k_2N)$ $= F(u+k_1,v+k_2N)$ $f(x,y) = f(x+k_1M,y) = f(x,y+k_2N)$ $= f(x+k_1M,y+k_2N)$
9)	Convolution	$(f \star h)(x,y) = \sum_{m=0}^{M-1}\sum_{n=0}^{N-1} f(m,n)h(x-m,y-n)$
10)	Correlation	$(f \star h)(x,y) = \sum_{m=0}^{M-1}\sum_{n=0}^{N-1} f^*(m,n)h(x+m,y+n)$
11)	Separability	The 2-D DFT can be computed by computing 1-D DFT transforms along the rows (columns) of the image, followed by 1-D transforms along the columns (rows) of the result. See Section 4.11.
12)	Obtaining the IDFT using a DFT algorithm	$MNf^*(x,y) = \sum_{u=0}^{M-1}\sum_{v=0}^{N-1} F^*(u,v)e^{-j2\pi(ux/M+vy/N)}$ This equation indicates that inputting $F^*(u,v)$ into an algorithm that computes the forward transform (right side of above equation) yields $MNf^*(x,y)$. Taking the complex conjugate and dividing by MN gives the desired inverse. See Section 4.11.

TABLE 4.4
Summmary of DFT pairs. The closed-form expressions in 12 and 13 are valid only for continuous variables. They can be used with discrete variables by sampling the continuous expressions.

	Name	DFT Pairs
1)	Symmetry properties	See Table 4.1
2)	Linearity	$af_1(x,y) + bf_2(x,y) \Leftrightarrow aF_1(u,v) + bF_2(u,v)$
3)	Translation (general)	$f(x,y)e^{j2\pi(u_0x/M + v_0y/N)} \Leftrightarrow F(u-u_0, v-v_0)$ $f(x-x_0, y-y_0) \Leftrightarrow F(u,v)e^{-j2\pi(ux_0/M + vy_0/N)}$
4)	Translation to center of the frequency rectangle, $(M/2, N/2)$	$f(x,y)(-1)^{x+y} \Leftrightarrow F(u-M/2, v-N/2)$ $f(x-M/2, y-N/2) \Leftrightarrow F(u,v)(-1)^{u+v}$
5)	Rotation	$f(r, \theta+\theta_0) \Leftrightarrow F(\omega, \varphi+\theta_0)$ $r = \sqrt{x^2+y^2} \quad \theta = \tan^{-1}(y/x) \quad \omega = \sqrt{u^2+v^2} \quad \varphi = \tan^{-1}(v/u)$
6)	Convolution theorem†	$f \star h(x,y) \Leftrightarrow (F \bullet H)(u,v)$ $(f \bullet h)(x,y) \Leftrightarrow (1/MN)[(F \star H)(u,v)]$
7)	Correlation theorem†	$(f \star h)(x,y) \Leftrightarrow (F^* \bullet H)(u,v)$ $(f^* \bullet h)(x,y) \Leftrightarrow (1/MN)[(F \star H)(u,v)]$
8)	Discrete unit impulse	$\delta(x,y) \Leftrightarrow 1$ $1 \Leftrightarrow MN\delta(u,v)$
9)	Rectangle	$\text{rec}[a,b] \Leftrightarrow ab\frac{\sin(\pi ua)}{(\pi ua)}\frac{\sin(\pi vb)}{(\pi vb)}e^{-j\pi(ua+vb)}$
10)	Sine	$\sin(2\pi u_0x/M + 2\pi v_0y/N) \Leftrightarrow \frac{jMN}{2}[\delta(u+u_0, v+v_0) - \delta(u-u_0, v-v_0)]$
11)	Cosine	$\cos(2\pi u_0x/M + 2\pi v_0y/N) \Leftrightarrow \frac{1}{2}[\delta(u+u_0, v+v_0) + \delta(u-u_0, v-v_0)]$
	The following Fourier transform pairs are derivable only for continuous variables, denoted as before by t and z for spatial variables and by μ and ν for frequency variables. These results can be used for DFT work by sampling the continuous forms.	
12)	Differentiation (the expressions on the right assume that $f(\pm\infty, \pm\infty) = 0$.)	$\left(\frac{\partial}{\partial t}\right)^m\left(\frac{\partial}{\partial z}\right)^n f(t,z) \Leftrightarrow (j2\pi\mu)^m(j2\pi\nu)^n F(\mu,\nu)$ $\frac{\partial^m f(t,z)}{\partial t^m} \Leftrightarrow (j2\pi\mu)^m F(\mu,\nu); \quad \frac{\partial^n f(t,z)}{\partial z^n} \Leftrightarrow (j2\pi\nu)^n F(\mu,\nu)$
13)	Gaussian	$A2\pi\sigma^2 e^{-2\pi^2\sigma^2(t^2+z^2)} \Leftrightarrow Ae^{-(\mu^2+\nu^2)/2\sigma^2}$ (A is a constant)

†Assumes that $f(x,y)$ and $h(x,y)$ have been properly padded. Convolution is associative, commutative, and distributive. Correlation is distributive (see Table 3.5). The products are elementwise products (see Section 2.6).

	Spatial Domain†		Frequency Domain†
1)	$f(x, y)$ real	$\Leftrightarrow$	$F^*(u, v) = F(-u, -v)$
2)	$f(x, y)$ imaginary	$\Leftrightarrow$	$F^*(-u, -v) = -F(u, v)$
3)	$f(x, y)$ real	$\Leftrightarrow$	$R(u, v)$ even; $I(u, v)$ odd
4)	$f(x, y)$ imaginary	$\Leftrightarrow$	$R(u, v)$ odd; $I(u, v)$ even
5)	$f(-x, -y)$ real	$\Leftrightarrow$	$F^*(u, v)$ complex
6)	$f(-x, -y)$ complex	$\Leftrightarrow$	$F(-u, -v)$ complex
7)	$f^*(x, y)$ complex	$\Leftrightarrow$	$F^*(-u - v)$ complex
8)	$f(x, y)$ real and even	$\Leftrightarrow$	$F(u, v)$ real and even
9)	$f(x, y)$ real and odd	$\Leftrightarrow$	$F(u, v)$ imaginary and odd
10)	$f(x, y)$ imaginary and even	$\Leftrightarrow$	$F(u, v)$ imaginary and even
11)	$f(x, y)$ imaginary and odd	$\Leftrightarrow$	$F(u, v)$ real and odd
12)	$f(x, y)$ complex and even	$\Leftrightarrow$	$F(u, v)$ complex and even
13)	$f(x, y)$ complex and odd	$\Leftrightarrow$	$F(u, v)$ complex and odd

Table 4.1

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